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AUTOMATIC BACKGROUND PERSON REMOVAL

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A thesis submitted in partial fulfillment of the requirements for the degree of "Bachelor of Science in Computer Science & Engineering"

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Abstract

Background removal is an essential task in computer vision with applications in video editing, augmented reality, and virtual conferencing. The research presents a background removal approach that integrates YOLOv11, the Segment Anything Model (SAM), and the LaMa inpainting network. The proposed pipeline first employs YOLOv11 to detect and localize target objects in the background within the image. The detected regions are then refined by SAM, which generates precise segmentation masks for the objects of interest. Finally, the LaMa (Large Mask Inpainting) model, which is a generative AI model, is applied to inpaint and seamlessly reconstruct the background in the masked regions, effectively removing undesired objects while preserving scene consistency. This whole methodology was applied to custom images to evaluate performance on images and later deployed in a web-cam to evaluate its real-time performance in a webcam video. Experimental results demonstrate that the combined approach achieves nearly accurate object removal, high-quality background restoration, and quite efficient performance suitable for real-time applications. The output demonstration for the video is in this link: https://youtu.be/VZZSa0ezUM4

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CHAPTER I

Introduction

1.1 Introduction

The ability to automatically remove background persons from camera frames in real-time presents a significant advancement for various applications, ranging from enhanced video conferencing to augmented reality [1]. This technology also offers substantial benefits in surveillance systems, where privacy concerns can be mitigated by anonymizing individuals not central to the monitoring objective, and in subject repositioning tasks, where dynamic elements need to be isolated or replaced [2]. The dynamic element can be removed using the image inpainting approach.

Image inpainting is a task of restoration of the missing region in an image. It is an important problem in Computer Vision and essential functionality in many image and in the graphical applications particularly object removal tasks. Image inpainting aims to remove unwanted content such as text, logo, timestamp, or even an entire person from an image while generating plausible backgrounds that blends seamlessly with the surrounding context. This makes the completed image look semantically consistent and realistic. Object removal using inpainting technique is widely applied in photography, film editing, digital restoration and image manipulation. It remains an active research area in image processing. Image restoration is studied by many studies and several methods and algorithms has been proposed [3].

Most of these methods are focused on inpainting objects in static positions and with particular sizes. But in recent years of studies deep learning has been added to the study of image restoration and many breakthrough results have been obtained, all of which benefit from the big data training [4]. But not all these methods can not be employed properly to inpaint object from the camera feed in real time due to the dynamism of the object. It is required to capture the object automatically properly in real time, segment it properly for better inpainting output and then inpaint it using any deep learning based approach.

This paper describes a hybrid pipeline that employs the latest YOLOv11 framework to capture background persons from the camera feed or video feed and separate foreground and background persons, Segment Anything Model (SAM) to segment the background objects, and the deep learning based Large Mask Inpainting (LaMa) model to remove the object and inpaint it with the background coherently.

1.2 Problem Statement

The emergence of advanced surveillance and security applications has accentuated the importance of efficient and accurate background removal, necessary for the identification of human subjects in complex and dynamic environments present in camera feeds [5]. Real-time performance is an issue with existing methods, especially when processing is to be done in real time under different illumination, occlusion, and the requirement of robust feature extraction for accurate separation of human bodies from the background [6]. This challenge becomes further compounded by the necessity or the system to operate effectively across diverse operational contexts. This contexts starts from border security to public space monitoring which often requires the ability to detect human presence from significant distances or within cluttered visual fields [7].

Moreover, the integration of advanced computer vision techniques such as convolutional neural networks is very important for enhancing the accuracy and reliability of human detection and recognition. Still, it is difficult for widespread deployment to develop robust algorithms capable of distinguishing and isolating human subjects from background elements, even under suboptimal conditions.

1.3 Objectives

The primary objective of this research is to develop and implement a complex system framework that automatically detects background person in real time from camera frame either in dynamic or static environment, remove it from background and inpaint with background environment. The following objectives are outlined:

- **Development of a Pipeline that E*ff*ectively Work on Real Time Camera Frame** Development of a hybrid pipeline that can be easily deployed on the Camera to do real time performance
- **Detecting Person E*A*ciently From Camera Frame:** Development of a component that effectively detect person in real time on each frame of the camera feed and effectively separate it from other objects in the background.
- **Utilization of Algorithm that E*ff*ectively Separate Foreground Person and Background Person:** For effective background person removal, an algorithm need to be derived and used to effectively seclude foreground person from background persons during inpainting operations.
- **Investigation of Real-World Inpainting Techniques:** Analyzation different inpainting techniques and learn how they can be applied on real world scenarios.

- **Improving Inpainting Quality and EAciency:** Leveraging the YOLOv11's accuracy in identifying regions requiring inpainting LaMa's ability to generate contextually relevant content and improve both quality and speed of the inpainting process [8].
- **Evaluation of LAMA Model's Performance:** Examination of the effectiveness of LaMa model in inpainting person from image on different complex environment and background. Evaluation o Model's performance using PSNR and SSNR parameters.
- **Providing a Robust, Adaptable, and EAcient Pipeline:** Implementation of a pipeline with combined strengths of YOLOv11, SAM and LaMa to offer a solution not only robust and accurate but also adaptable to a wide range of image processing needs.

1.4 Scope

The author used different modern tools and libraries to carry out the research work. The modern tools that are used for the work are:

- **YOLOv11 (You Only Look Once, Version 11)**

  YOLOv11 is a modern and real-time object detection system that is known for its speed and high accuracy. It is used in work in purpose to accurately identify and localize objects within images, videos and generate precise masks for the regions requiring inpainting. YOLOv11's streamlined, single-step prediction of the bounding boxes and object class enables it to efficiently detect even in complex scenes. Its backbone architecture and use of anchor boxes make it suitable for varied applications such as self-driving cars, surveillance, and medical imaging.

- **SAM (Segment Anything Model)**

  The Segment Anything Model is another advanced model that is employed to segment object with minimal human input. It makes the process highly versatile and interactive. It is a promptable tool that is employed to segment with human part from the bounding box generated using YOLO model to effectively separate it from the background of the object. The Image Encoder(ViT) is used for the effective operation of this tool.

- **LaMa (Large Mask Aware Inpainting)**

  LaMa is an advanced image inpainting model that is designed to handle large missing areas and complex geometry [9]. It leverages Fast Fourier Convolutions (FFC) that allows the model to capture global context of the image efficiently which is crucial for image inpainting [9]. LaMa processes masked images and blends the filled regions seamlessly with the original image, maintaining high visual quality even for high-resolution inputs.

The pipeline of the author integrates YOLOv11 to generate high precision masks delineating areas for inpainting. Using the bounding box, SAM synthesizes better masked images. Applying LaMa The masked maps and the mask information are then input to LaMa for the map reconstruction. The incorporation of these contemporary tools greatly advances inpainting quality, speed, and versatility in diverse application domains such as restoration, content creation, and medical imaging.

1.5 Unfamiliarity of the Problem

The background object removal problem is discussed in many works in recent times. It is one of the versatile application of image inpainting technique. But applying the background object removal in real time from the camera frame efficiently and fastly still remains unexplored. The existing tools and methods struggles to capture object and automatically remove it and reconstruct background on each camera frame in both daylight and artificial light. The problem discovered for the local camera that detects any static and dynamic person in background on live stream camera feed and inpaint it with the background. Furthermore, there is limited research that combines object detection and segmentation with the generative deep learning model in comprehensive background object removal framework. The combination of different model strength, analysis of the performance of each model, derivation of algorithm to remove background persons and applying the overall pipeline on a live stream feed of local camera makes this problem particularly novel and technically challenging.

1.6 Contributions

This Contributions of the author in research works are:

- Collecting dataset of random images of persons in different environment and activity.
- Fine Tuning models for better performance on the dataset.
- Separating foreground person from background and generating masks on the background persons of the image.
- Combining the inpainting model with the object detection and mask generation framework for the most effective restoration of background for the removed object.
- Deploying the overall pipeline on the local camera of laptop device to ensure the real time performance of the ovrall pipeline.

1.7 Project Planning (In Weeks)

The following Gantt chart represents the projected aspiration of the overall progress of the proposed method in this report from the starting week to the week till pre-defence and then from the week after predeence to the week till defense.

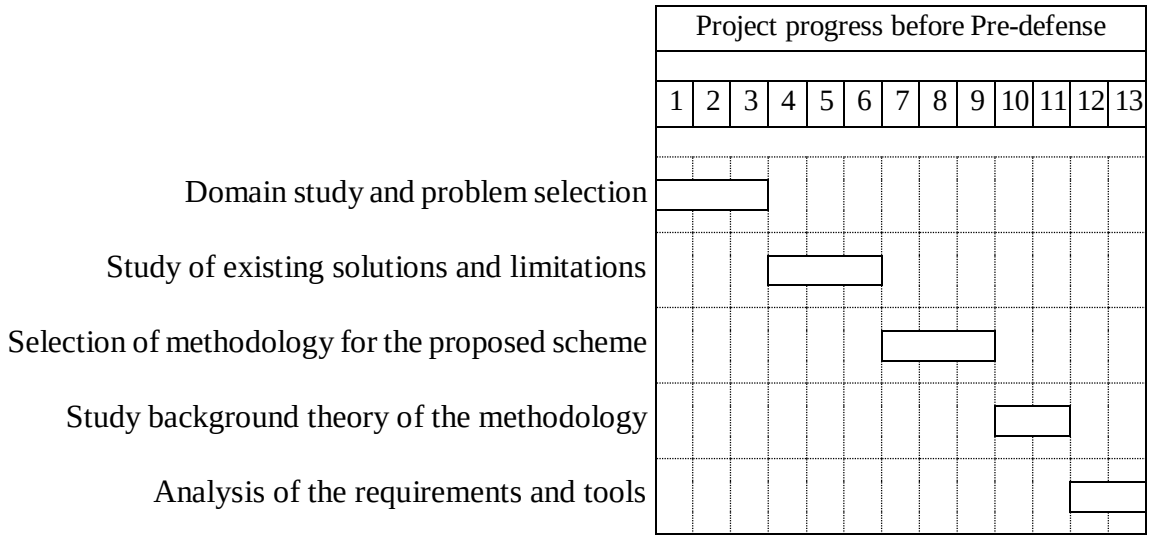


Figure 1.1: A Gantt chart showing the progress before Pre-defense

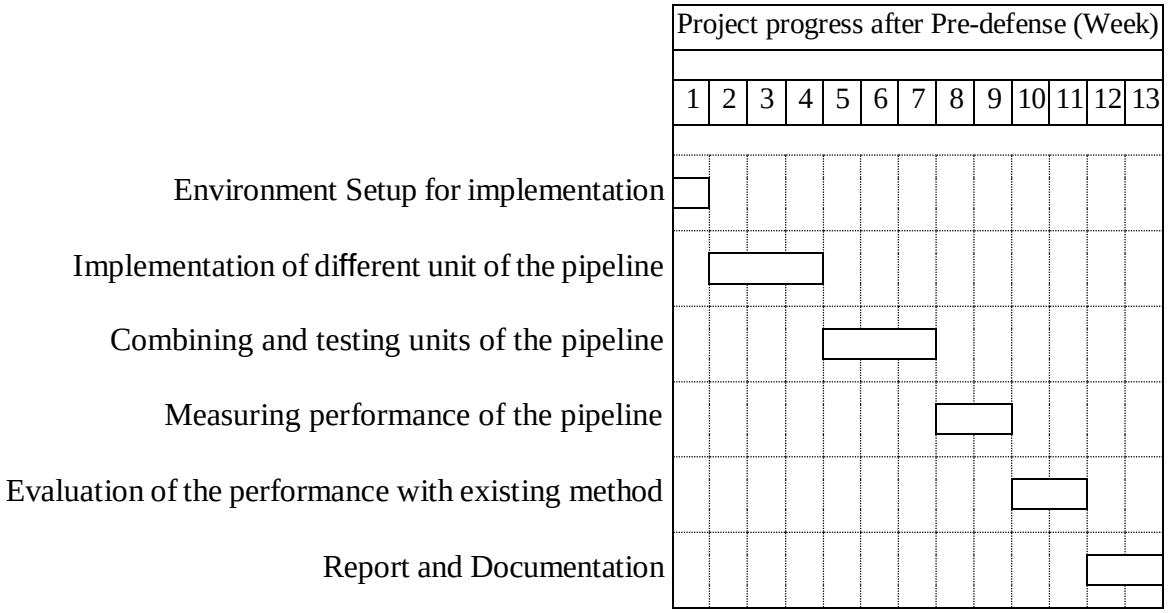


Figure 1.2: A Gantt chart showing the progress after Pre-defense

1.8 Applications of the Work

The findings from this research can have broad applications in particular scenarios requiring immediate analysis and response from the system. This includes:

- The proposed method can detect and inpaint unwanted or sensitive objects in live video feeds. It may include anonymizing faces in requirement of privacy or removing sensitive information immediately.
- The proposed method can help autonomous systems perceive their environment properly by filling occluded or missing data areas in sesnsor inputs. It may help improving navigation, obstacles avoidance and situational awareness.
- The proposed method can help in media production. Live video streams can have objects dynamically removed without interrupting the flow. It enables virtual advertising, background replacement or censoring inappropriate content in real time.
- The proposed method can play good role in medical imaging. It can perform enhancement and reconstruction of the medical video feeds such as endoscopy or ultrasound can provide improved visualization for the doctor.
- The proposed method can enhance AR/VR experiences by seamlessly integrating virtual objects into real world view. It can also remove real object obstructing the view.
- Live camera streams that monitor road conditions can identify potholes or objects, and apply inpainting to synthesize acenes that offer a cleaner view, fascilitating immediate maintainance actions and enhancing public safety.

1.9 Organization of the report

The rest of the thesis is organized as follows:

Chapter I Provides an introduction to the background, problem statement, objectives, and organization of the thesis.

Chapter II Presents short summaries of the related research works on real time automatic background person removal from camera frame.

Chapter III Presents the methodology including YOLOv11 model analysis, SAM model analysis,

Chapter IV Details the overall experimental setup and dataset description, description of different implementation steps, result analysis and evaluation of the results.

Chapter V Discusses Social, Economic impact of the work.

Chapter VI Discusses complex engineering activities associated with current thesis work.

Chapter VII Discusses challenges, future works, and conclusion.

CHAPTER II

Literature Review

2.1 Introduction

This chapter provides a comprehensive overview of the removal of objects in real time using object detection methods and image inpainting approach. The author examined different approaches of object especially person detection and remove it with the background using different inpainting technique. The existing methods advantages and their limitations are examined by the author and tried to overcome those limitations using the proposed method's pipeline. This section discusses different methods of real time framing of objects, object detection and object inpainting techniques. It also provides discussion of capable needs to overcome these existing methods limitations.

2.2 Literature Review

There has been many methods that are focused on real time framing of objects in camera, detecting this objects and different inpainting techniques related to removal of the objects. Background subtraction is a method that detects all motion objects from static background by comparing each frame with the background model[10]. A deep neural network is also combined with the background subtraction model or rapid detection of person and vehicle. But the method can only detect the motion of the object and can not classify actually what object moves. Also the method struggles when there is clotted or complex background.

Khanorkar et al. [11] provides a method of real time background removal. The method combines Mediapipe SelfieSegmentation, OpenCV and OBS Virtual Camera that works decently for lightweight and quick deployment. The Mediapipe's SelieSegmentation model is used to generate person background mask. The mask is then used to remove and replace the background with images stored in the local directory and accesed camera is the webcam of the local pc. There has been many inpainting based object removal method.

Sanchez et al. [12] has provided a method of real time inpainting that combines lightweight CNN for semantic structure at low resolution with a patch-based mechanism. This mechanism copies high-frequency texture from known regions. There has also been traditional inpainting method that are focused on object removal and semantic reconstruction of background.

Pathak et al. [13] proposed the use of context-encoder to fill in the missing region along with the generative network. He firstly train deep neural networks region of 64×64 in a

128×128 images with both pixel wise reconstruction loss and generative adversarial loss as the objective function. But it consider the structure of holes only at the training phase and not the inference stage of the image which ultimately produces distorted and blurry images when there is arbitrary shape of hole.

A different method was used in [14] that uses a patch offset to fill the missing region of image. This method uses basics characteristics of an image that is suitable for completing a large scene with less symmetrical pattern. So this method fails to work in the license plate and human face where there is abundant details. So it is difficult to complete image just using only the pattern recognition of the uncorrupted part of the image.

Pyo et al. [15] introduced a combined GAN based approach for object removal and inpainting. This method utilizes two GAN network that parallely works on removing an object from the background and replace it with background. One GAN network is utilized for object removal purpose and another one is used to perform the reconstruction by utilizing the generator and discriminator architecture. The generator reconstruct the missing part and discriminator compares it with the background. But main problem arises when the missing region is too large it sometimes fails and creates blurry artifacts.

The overall pipeline has similarity with [8] but they were not focused on real time camera deployment of the overall method. They just developed the pipeline combining YOLOv8 and LaMa model together for better efficiency on the inpainting process. However, their approach was primarily evaluated in offline or batch-processing scenarios, limiting its applicability for dynamic use cases such as surveillance, live streaming, or interactive applications.

Criminisi et al. [16] provides the method that combines both texture filling and inpainting method. They introduced an algorithm that targets the region to be removed and filled. The algorithm follows an iterative process untill all pixels in the target region are filled.

Barnes et al. [17] proposes a method that tries to find the most similar patch in the undamaged area to fill the image with that patch. It works well for large areas but there is a computational limitation that it can not give satisfactory result when the missing area has arbitrary shapes.

Tu et al. [18] proposed a variational autoencoder approach for image inpainting that displays improved reconstruction quality. It can learn complex features of image and can create smooth image used for the inpainting. But the result is not very well for all resolution images that are operated on.

Liu et al. [19] proposed a generative adversarial network-based method for image inpainting, achieving realistic image completions. They used the semantic inpainting method of the image to maintain consistency by employing the prior loss, gradient loss and reconstruction loss. But it only works on the fixed image with 64×64 pixel.

Mirkamali and Nagabhushan [20] proposed a depth-wise image inpainting method for object removal inpainting. The method combines the advantages of both exampler-based and depth-based algorithms. The exampler-based algorithm is used to fill holes using available patches

while depth-based methods distinguish foreground from background before inpainting. A depth map of the target image is used to determine the fill order of all objects within it.
Tuptewar and Pinjarkar [21] proposed a robust exemplar-based approach for image and video inpainting, enabling effective object removal and region filling. The approach includes robust segmentation of the image which is region based segmentation before exemplar based inpainting. The inpainting in performed based on the networking of two networks.
Wu and Ruan [22] proposed a cross-isotopes exemplar-based inpainting method to remove unwanted objects while preserving image structure. The paper follows isotopic method of inpainting and removal of object following the use of exampler-based method.
Chang et al. [23] proposed Vornet, a spatial-temporally consistent video inpainting framework for object removal. VORNet takes as input video frames with target bounding box masks and generates output video frames without the target object. It is built upon three networks which are wrapping networks, inpainting networks and refinement networks.
Yu et al. [24] introduced Inpaint Anything, which combines SAM with image inpainting. The paper represents a novel framework that integrates the capabilities of various foundation models to address a wide range of image inpainting problems. The authors employ a dual encoder architecture, processing RGB and depth images separately, and then fuse the encoded features using an attention mechanism in the decoder. This integration significantly improves the model's ability to reconstruct images with greater accuracy and contextual awareness Table 2.1 provides comparison of different important methods and their weakness.

Table 2.1: Comparative table of important related works and their weakness

Paper	**Publisher**	**Shortcomings**
Kim et al.	Springer	Sensitive to the lighting changes.
Pyo et al.	IEEE	Struggles more on reconstructing the global structure of the image.
Sanchez et al.	IEEE	Performs poor inpainting on large missing area.
Khanorkar et al.	Dialnet	Improves edge smoothness and overall mask quality.

2.3 Discussion of Research Gap Solution

The several existing scheme contains limitations. The limitations of some papers and how proposed method can solve these limitations are given in Table 2.2.

Table 2.2: Issues in the existing models and how the proposed method deals with them

Serial No.	Existing Methods	Issues	Proposed Solution in this Method
1	Hybrid framework combining background subtraction and deep neural networks for rapid person detection	Sensitive to lighting changes	Learns robust object features and can operate effectively under varying lighting conditions.
2	A new benchmark and baseline for real-time high-resolution image inpainting on edge devices	Struggles to reconstruct the global structure of images in diverse scenarios	Utilizes LaMa with FFC model, which performs well on global reconstruction of missing regions.
3	Object removal and inpainting from images using **GAN**-based approaches	Poor inpainting performance on small missing regions	Performs accurate inpainting even on large missing regions.
4	Real-time background removal in videos	Suffers from video quality degradation and produces unsmooth mask edges	Improves edge smoothness and overall mask quality.

CHAPTER III

Methodology

3.1 Introduction

In this chapter, the author describes the comprehensive methodology used to perform the background object detection, removal of the the object from the background and restoration of the background in real time environment. The authors approach involves combining the YoLOv11+SAM model with the FFC(Fast Fourier Convolution) based LaMa model and implementing and optimizing its performance in the camera frame for real time operation. The methodology encompasses the separate models implementation and training, rigorous algorithm analysis for the implementation steps and analysis o the basic frameworks used in the hybrid pipeline.

3.2 Overview of Overall Methodology

The authors research employs use of three different models which are YOLO model for perfectly capturing background person, Segment Anything Model to properly segment the person from bounding box, mask generation procedure, and lastly utilizing the LaMa model to remove person in masked region and reconstruct that part to seamlessly blend with background. Figure 3.1 describes the overall workflow of authors. The figure is given bellow:

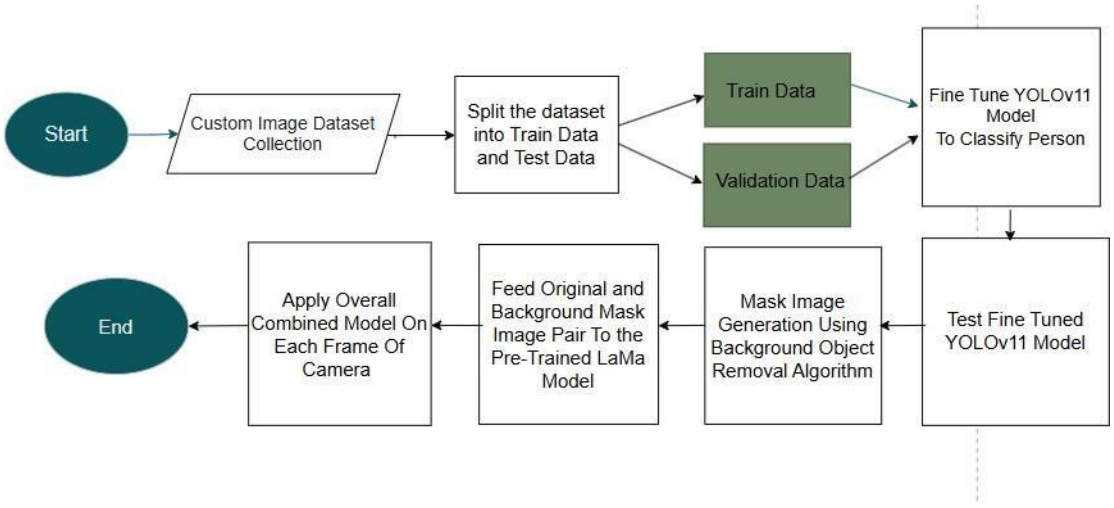


Figure 3.1: Overall Workflow of the Proposed Method

3.3 Description of Di*ff*erent Steps of Proposed Method

The proposed method is implemented utilizing different model frameworks. The objective was to develop a pipeline that will efficiently work on the real time camera frame.

3.4 Custom Image Dataset Collection

The first step of implementing the proposed method is the dataset collection. The dataset need in the paradigm to train the YOLOv11 model is the custom roboflow dataset. The dataset contains images of persons with different complex background. The collected images would be of different sizes. The collected images are then resized to 640 $\times$ 640 pixels. Then the dataset is split into two parts which are trained dataset and validation dataset. The rest of the images are test images used to evaluate the performance of the finetuned YOLO model. The person images were labeled in roboflow. To ensure consistency, the author need to perform different data augmentation techniques such as rotation, flipping, and scaling, enhancing the model's robustness to different input variations.

3.5 YOLOv11 Model Analysis

A deep neural network architecture based on the Darknet Framework is used by YOLOv11. The convolutional layer makes up the architecture which produces a feature map preserving spatial data. In the performance of YOLO model the image is divided into grid cells. Each grid cells in the image is in charge of the forecast of bounding boxes and class probabilities. YOLOv11 is known as grid-based object detection system that utilizes all these cells to forecast bounding boxes and class scores. YOLOv11 uses anchor boxes to enhance the prediction of the bounding box. These anchor box predicts precise bounding box dimensions for various item scales and aspect ratios. The mathematical representation of the detection process can be described as:

$$\text{Conv1} = \text{Convolution}(X, W_1) + B_1 \tag{3.1}$$

$$\text{ReLU1} = \text{ReLU}(\text{Conv1}) \tag{3.2}$$

$$\text{Conv2} = \text{Convolution}(\text{ReLU1}, W_2) + B_2 \tag{3.3}$$

$$\text{ReLU2} = \text{ReLU}(\text{Conv2}) \tag{3.4}$$

$$\text{Detection} = \text{OutputLayer}(\text{ReLUfinal}) \quad (3.5)$$

Applying filter across the pixel grid is the operation that is known as convolution. The filter is considered as a tiny window that moves across the picture and adds up the weights of the underlying pixels. The process co-operates in the extraction of the image's edges, textures, and patterns. Equations 3.1 and 3.3 represent this convolution process. Each convolution operation adds the bias terms (B1 and B2) as constant values to the output. The operations aid in modifying the neurons' general level of activity in the following layer. The model becomes non-linear because to the Rectified Linear Unit (ReLU) activation function. The rectified linear unit merely applies a threshold to the convolution's output and maintains the positive values constant while setting negative values to zero. Along with detecting objects, YOLOv11 also classifies them. Each of the bounding box prediction includes a probability distribution over different classes that allows the model to identify and label multiple objects in the image simultaneously.

The architecture of YOLOv11 is designed to optimize both speed and accuracy, building on the advancements that is introduced in earlier YOLO versions like YOLOv8, YOLOv9, and YOLOv10. The main architectural innovations in YOLOv11 revolve around the C3K2 block, the SPFF module, and the C2PSA block, all of which enhance its ability to process spatial information while maintaining high-speed inference.

YOLOv11 uses C3K2 blocks to handle feature extraction at different stages of the backbone. The smaller 3x3 kernels that are used allow for more efficient computation while retaining the model's ability to capture essential features in the image. At the heart of YOLOv11's backbone is the C3K2 block, which is an evolution of the CSP (Cross Stage Partial) bottleneck introduced in earlier versions. The C3K2 block optimizes the flow of information through the network by splitting the feature map and applying a series of smaller kernel convolutions (3x3), which are faster and computationally cheaper than larger kernel convolutions. By processing smaller separate feature maps and merging them after several convolutions, the C3K2 block improves feature representation with fewer parameters compared to the C2f blocks of YOLOv8.

YOLOv11 retains the SPFF(Spatial Pyramid Pooling Fast) module designed to pool features from different regions of an image at different scales which improves the network ability to capture object of different size. It is truly a new feature of YOLOv11 version which was not found in previous version. It performs multiple max pooling operation and later aggregate the multi-scale contexual information. The inclusion of SPFF ensures that YOLOv11 can maintain real-time speed while enhancing its ability to detect objects across multiple scales.

YOLOv11 add C2PSA (Cross Stage Partial with Spatial Attention) block to make the model focusing on the important spatial regions in the image. Figure 3.2 gives the overall description of the architecture of YOLOv11 model collected from official site of YOLO:

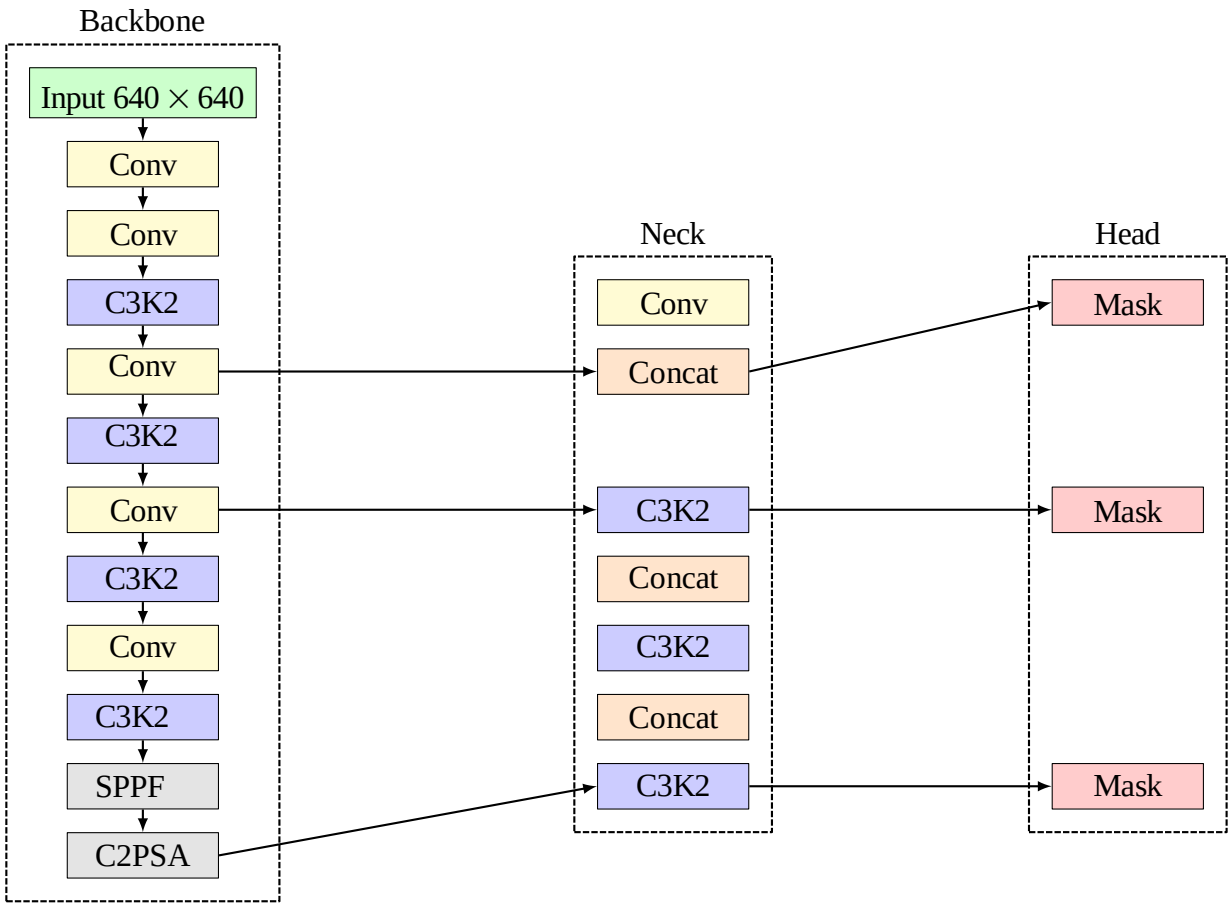


Figure 3.2: Simplified YOLOv11 architecture showing Backbone, Neck, and Head

3.6 Segment Anything Model Analysis

The image is then fed to the ViT (Vision Transformer) based Segment Anything Model. The Segment Anything Model takes as input the image and also the prompt. The prompt tell which region to inpaint. The encoder processes the image once and produces a dense embedding (feature map) that represents the entire image. The prompts are also converted to embeddings. All the embeddings are lightweight. They are added to the image embedding. A lightweight transformer-based decoder takes the combined-input(image+prompt). It outputs one or more binary masks with confidence score. The Segment Anything Model finally returns the segmentation masks for the object person within the bounding box. The figure 3.2 shows the overall methodology of Segment Anything Model.

3.7 Dual Background Object Removal Algorithm Analysis

The procedure follows with detecting the background object with YOLOv11 model. The YOLOv11 model does not only detect the object but also classifies them. The bounding box prediction includes probability distribution over different classes. It allows the model to

identify and label multiple objects in the image simultaneously. YOLO makes all the detection and keeps the person (class id=0). For the background object removal two algorithm is proposed. The algorithms can be used to separate the foreground person and background person for background object removal purpose. The first algorithm is based on the bounding box mask area calculation. From the YOLOv11 bounding box the area for all person is calculated. Then the person in bounding box with largest area is considered the fore ground person and rest is considered the back ground person. The second algorithm is based on the comparison of bottom Y-coordinate. The person with the lowest Y-coordinate is considered the foreground person. So the bounding boxes are sorted based on the bottom Y-coordinate and foreground background persons are separated for the mask generation approach. Later the mask image can be generated for the background person in order to remove them from the context of the image. Either of the two approach can be applied based on the better inpainting quality. The pseudocode for the first algorithm is given bellow:

Algorithm 1: Foreground–Background Separation (Bounding Box Area Rule)

Data: Frame I with detected objects and bounding boxes from YOLO
Result: Binary mask M highlighting background persons
Extract all bounding boxes $B = \{b_1, b_2, \ldots, b_n\}$ where class = "person"
if $n \leq 1$ **then**
return empty mask M **else**
Compute area $A_i = (x^i_2 - x^i_1)(y^i_2 - y^i_1)$ for each $b_i \in B$
Foreground person $\leftarrow$ box with largest area $\max(A_i)$
Background persons $\leftarrow B \setminus \{\text{Foreground}\}$
Initialize empty binary mask M **for** *each background person box* b_j **do**
Fill M with 255 inside coordinates of b_j
return M

In the pseudocode of the algorithm the (x1,y1) and (x2,y2) are the co-ordinates of the bounding box used to calculate the area of the each bounding box. The pseudocode for the second algorithm is given bellow:

Algorithm 2: Foreground–Background Separation (Lowest Y Rule)

Data: Frame I with detected objects and bounding boxes from YOLO **Result:** Foreground mask M_f and Background mask M_b

Detect all person bounding boxes $B = \{b_1, b_2, \ldots, b_n\}$

if $n < 2$ **then**

return empty masks M_f, M_b **else**

Sort bounding boxes B by bottom coordinate y_2

Foreground person ← box with maximum y_2 (lowest position)

Background persons ← $B \setminus \{\text{Foreground}\}$

Apply SAM on background boxes to obtain masks M_b

Output foreground mask M_f and background mask M_b

The whole procedure follows the overall proper mask generation process. The python OpenCV takes the result from SAM and then it converts it to numpy values which are set to 255 for 1 and 0 for 0 values. The final mask is a binary image with 255 (White) for the person to be removed and and 0 (Black) for everything else to be preserved.In addition, resizing and morphological operations (such as dilation or smoothing) can be applied to refine the mask boundaries, reducing artifacts and ensuring that the separation between foreground and background is accurate. The resulting mask serves as the critical input for the LaMa inpainting model, allowing it to seamlessly fill in the removed regions with contextually consistent background information. We can consider the overall mask images as the function of M, M prime as the scale mask values, 255 for the scaling factor and x as the original Pixel location. Furthermore, advanced morphological operations like erosion and opening can be applied iteratively to remove small noise and smooth the mask edges for cleaner segmentation. This refinement helps prevent bleeding artifacts during inpainting, especially around fine details such as hair or clothing edges. The binary mask's pixel values act as a strong signal for the LaMa model, guiding it precisely where to reconstruct missing visual information. The overall equation for the Mask image generation:

$$M'(x) = 255 \cdot M(x) \tag{3.6}$$

Table 3.1: Parameters of mask transformation

Parameter	Description
$M(x)$	Original hole mask (binary: 0 or 1)
$M'(x)$	Scaled mask values (0 or 255)
255	Scaling factor used to convert binary mask into intensity values
x	Pixel location

3.8 LaMa Model Analysis

LaMa takes the output masked image which is accompanied by the mask as a tensor as a tensor with 4 channels and produces the resulting RGB image in a completely convolutional approach. It is important for the inpainting models to have access to the global context as soon as possible. If not the generator might observe only the regions containing the only the missing pixels. It would result in wastage of computational resources to build up the context over a series of layers. LaMa deals with this problem with fully differentiable Fast Fourier Convolutions [26]. It is a recently proposed layer that process input in two branches: a local branch that uses the standard convolutions, and a global branch that uses a channel-wise real FFT to transform the input into the frequency domain. Then it performs 1x1 convolution to the frequency domain, and recovers spatial structure with real IFFT. The these two outputs of two different branches are concatenated channelwise. The application of fast Fourier Convolutions presents two significant advantages: they provide the inpainting model immediate access to the global context of the image and effectively capture periodic patterns frequently seen in pictures (e.g., brick structures, steps, etc.). An algorithm is described in [8]. Fast Fourier Convolution makes following steps:

Algorithm 3: Fast Fourier Convolution Algorithm

Data: Input tensor X with dimensions $\mathrm{R}^{H\times W\times C}$
Result: Output tensor Y with dimensions $\mathrm{R}^{H\times W\times C}$
Apply Real FFT2d to X: $X_{\text{fft}} \leftarrow \text{RealFFT2d}(X)$;
Concatenate real and imaginary parts:
$X_{\text{concat}} \leftarrow \text{ComplexToReal}(X_{\text{fft}})$;
Apply convolution block in the frequency domain:
$Y_{\text{fft}} \leftarrow \text{ConvBlock}(X_{\text{concat}})$;
Apply inverse transform to recover spatial structure:
$Y_{\text{concat}} \leftarrow \text{InverseRealFFT2d}(Y_{\text{fft}})$;
Recover final output tensor:
$Y \leftarrow \text{RealToComplex}(Y_{\text{concat}})$;
return Y;

This algorithm performs a 2D real-valued Fast Fourier Transform (FFT) on the input tensor X to compute the frequency domain. It disjoins and appends the real and imaginary parts, and employs a convolution block in the frequency domain to deal with this representation. It applies INVRFFT, and then applies the 2D real fft to transform to the spatial domain as before, reconstructing the output tensor Y from the complex result that will be returned. This allows convolution to be performed relatively easily in the frequency domain. This algorithm exploit the computational efficiency of frequency domain convolution, which can be much less expensive than spatial convolution. The convolution block also works on concatenated real and imaginary part of the input tensor by shifting the input tensor into the frequency

domain, and thus could model complex patterns. Lastly, the inverse FFT reconstructs the spatial representation using the transformed features, ensuring the original tensor dimensions remain the same while augmenting or otherwise transforming the input data. The following figure describes the overall workflow of the FFC based LaMa model taken from [9]:

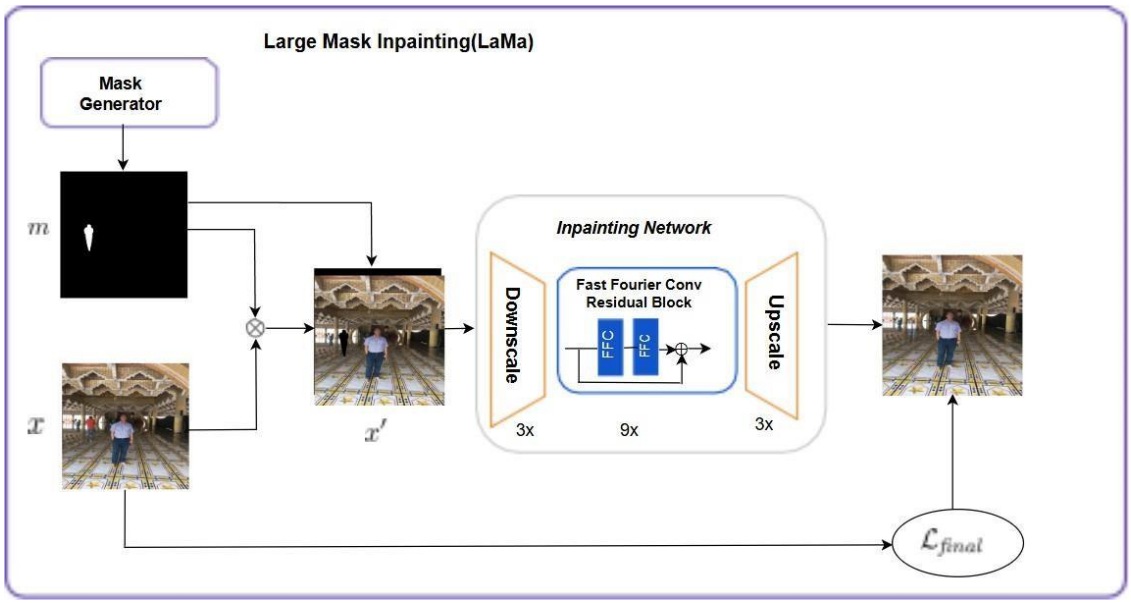


Figure 3.3: LaMa FFCbased Framework Working Overview[9]

The whole procedure of Fast Fourier Convolution can be briefed in the eqn:

$$F(u, v) = \sum_{x} \sum_{y} f(x, y) \cdot \exp^{\cdot} - i\, 2\pi(ux + vy) \tag{3.7}$$

Table 3.2: Description of terms in the Fast Fourier Transform equation (Eq. 2)

Symbol	Description
$F(u, v)$	Frequency domain representation at frequency coordinates (u, v); contribution of spatial pixels to that frequency component.
$f(x, y)$	Intensity value of the pixel at spatial location (x, y) in the image.
(u, v)	Frequency coordinates corresponding to horizontal (u) and vertical (v) frequency components.

For every pixel in the image, the FFT performs multiplication on the pixel value (f(x, y)) using the complex exponential function then totals these items for every potential frequency

element (u, v). The outcome is a novel value within the frequency domain (F(u,v)) that indicates the role of that particular pixel in a specific frequency element in the image. LaMa inpainting architecture is developed like the resnet architecture[25].

There are 3 downsampling blocks, 6-18 residual block and 3 upsampling blocks. The inpainting network of the LaMa model is similar to U-NET–like architecture with downsampling and upsampling arms and the core part between them with a sequence of FFC Residual Blocks (up to 18 in the Big LaMa model). All FFC is implemented properly within residual block. The main architecture of LaMa model is a mix of generator and discriminator where generator part is the main part of the architecture that performs the inpainting of the masked part of the image is the LaMa FFC based generator. An additional patch discriminator is attempted to be added by the author for making the inpainting more realistic. The Fast Fourier Convolution used in the Framework is a mix of local mix of local features (extracted through standard convolutional layers) added to global features (from a frequency domain).

It works on high receiptive field that allows the Convolution operation to extract global features from the overall images. The key component that captures global features is the Spectral Transform block where channel-wise FFT (Fast Fourier Transform) is applied. Then it performs a 1x1 convolution in the frequency domain, and finally recover spatial structure with inversed FFT. The one major advantages of this is that even though the overall framework was trained on low resolution images like image with sizes of 256×256 or 512×512 it still works well with high resolution image frame. The figure bellow represents the overview of the overall work of the LaMa FFC based framework.

LaMa utilizes the dilated convolutions in some layers to enlarge receptive field without additional parameters, which further helps LaMa to fill in the large missing regions. Texture repetition is also not over utilized in the latest architecture, the emphasis stays on semantic coherence throughout the image. Yet another strength is that it is efficient—in spite of its strong global feature extraction, LaMa does not lose its lightweightness in terms of computational complexity with respect to other GAN-based inpainting models. This makes it more appropriate for real or near real time application. In addition, the framework shows robustness to complex and irregular masks which has been a weakness in previous inpainting methods. Its frequency-based representation makes it more generalizable to other domains with sharper and more cohesive reconstructions, even in hold-out contexts.

Finally, LaMa's architecture can provide a starting point which can then be extended by integrating with attention mechanisms or multi-scale discriminators for better realism and detail preservation. The FFC operation is the added to res-net like architecture with 6-18 residual block. The conventional model like ResNet[25] actually do not work well for the full growth of effective receptive field[27]. Receptive field might not suffice especially the early layers of the network,due to the typically small sized convolutional kernels.

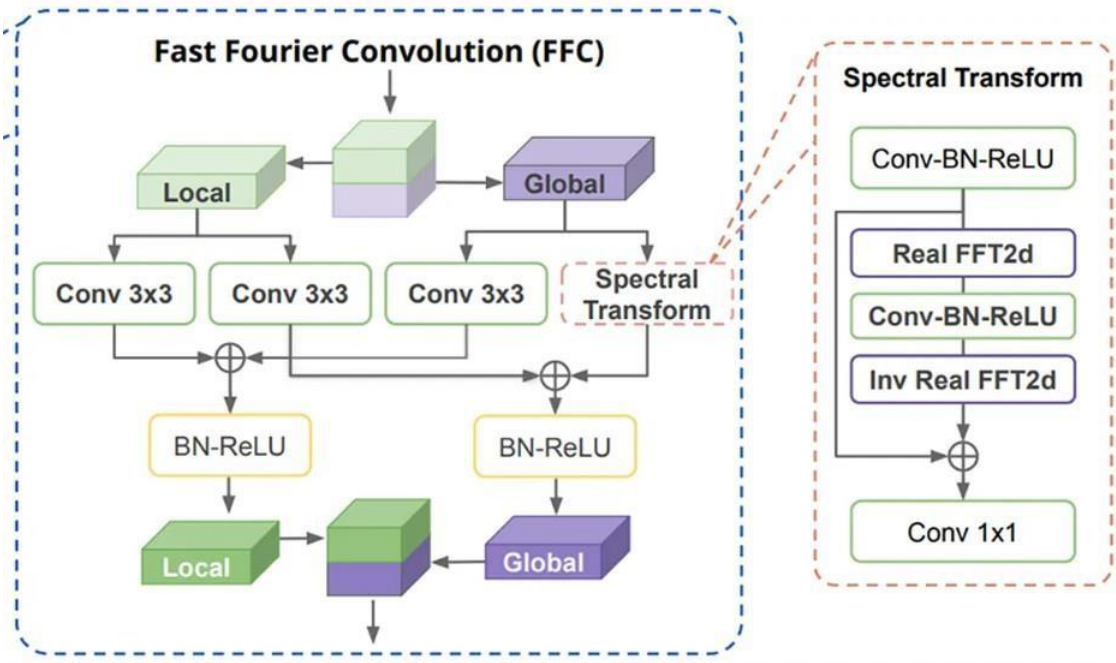


Figure 3.4: Fast Fourier Convolution (FFC) block diagram Overview[9]

For this reason the global context will be lacking result in the computational wastage of computations and parameters to create one. After applying FFC it works on the global context of the image. It is applied on the residual block.The residual block is the common component of the resnet-like architecture. Each convolutional layer is typically followed by batch normalization and a ReLU activation function. The input x to the block is fed through these layers to produce a transformed output. The original input x simultaneously bypasses the convolutions via a skip connection or identity connection. The output of the block is the element-wise sum of the transformed output and the input.

If dimensions differ between x and F(x) a 1x1 convolution is applied on to match dimensions before addition. This addition is typically followed by a ReLU activation. This structure helps mitigate vanishing gradient problems by enabling easier information and gradient flow through deep networks. In deeper networks, residual blocks also enable modular stacking, allowing architectures like LaMa to incorporate a large number of layers without significant degradation in performance. Additionally, the skip connections act as implicit regularizers, reducing overfitting and improving generalization across datasets.

Residual learning further encourages faster convergence during training, as the optimization process focuses only on refining residual mappings rather than relearning entire transformations. Moreover, the structure naturally supports parallelization, making the computation more efficient on modern GPUs. It allows the block to learn residual functions instead of the full mapping which makes training deep networks more effective. With regard to the LaMa framework, FFC into residual blocks increases their capability of capturing

global dependencies, which the standard convolutions cannot learn effectively since their receptive fields are narrow. This cocktail of features endows the residual block the ability to learn and handle texture of local information and structure information of distant ranges. In this way, the inpainting model is not only able to fill in missing areas more coherently and with large spatial contexts. In addition, the residual design also helps to keep training stable for global feature extraction via FFC for different input overall resolutions. Residual blocks also enhance model interpretability, as the skip connection provides a clear path for analyzing how much transformation is learned versus preserved from the original signal. This makes it easier to study layer contributions during ablation experiments.

Another strength lies in their robustness: residual architectures are less prone to degradation in performance when scaled to very deep networks, making them well-suited for large-scale inpainting tasks where capturing both fine-grained details and broad contextual structures is essential. The residual design also contributes to better numerical stability during training by maintaining a smoother loss landscape, which reduces the likelihood of exploding or vanishing gradients. In the LaMa framework specifically, this modular design allows seamless integration of frequency-domain enhancements such as FFC, enabling the network to model both spatial and global semantic consistency. Furthermore, residual connections facilitate the reuse of low-level and intermediate features throughout the network, helping preserve important structural information while deeper layers focus on higher-level representations. This is particularly beneficial for inpainting, where the network must maintain continuity in textures, edges, and object boundaries across the missing regions. The combination of residual learning and large receptive fields enables LaMa to infer missing content from distant contextual information rather than relying solely on neighboring pixels. Consequently, the architecture can produce more coherent and visually consistent restorations, especially when the missing regions are large or contain complex structures. Moreover, residual blocks allow the network to learn the difference between the corrupted input and the desired complete image, simplifying the overall reconstruction process. This helps the model concentrate on recovering missing structures rather than relearning information that is already present in the input. The architecture is also capable of handling irregularly shaped and relatively large missing regions while preserving the surrounding image context. By combining local feature extraction with global information aggregation, LaMa can generate realistic textures and maintain structural continuity across the inpainted area. These characteristics make the residual-based design particularly effective for high-resolution image restoration and real-time or large-scale image inpainting applications. The figure 3.4 shows the general architecture of the residual block of LaMa:

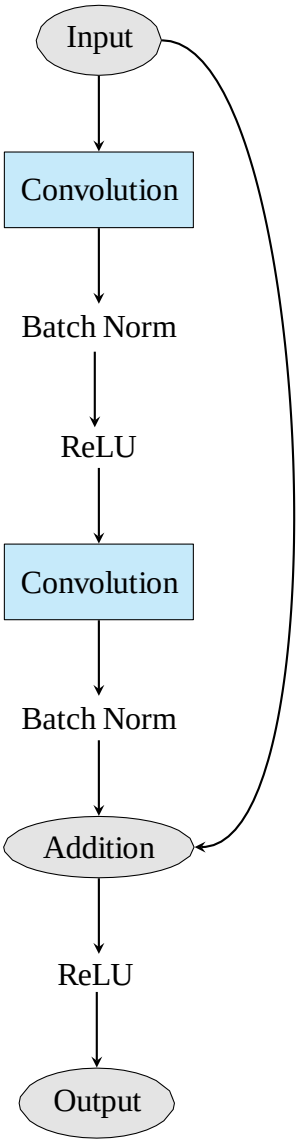


Figure 3.5: Flowchart illustrating the convolutional block with skip connection including convolution, batch normalization, ReLU activation, and addition[9]

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In the general architecture of the residual block the convolution al layers are replaced as the fast Fourier based convolutional layer. The fast fourier based convolution al layer works on the global context of the image. It extracts the global feature of the image which describes the whole information of the image to fill the masked region of the image. It maintains the coherency of the image. Later a patch discriminator is added to LaMa to make it a completely generator discriminator style architecture framework that performs background reconstruction is done very very smoothly.

3.9 Deployment and Application of the Whole Pipeline to Local Camera

After the implementation and performance analysis of the each framework of the pipeline the whole pipeline was deployed in a camera for real time operation. The main goal of the proposed method is to implement the system in real time. To perform this whole deployment procedure the author used mainly YOLOv11 and LaMa model combined on each frame of the webcam. The webcam each frame is then processed with the parallel YOLOv11 model and LaMa model to perform the background object removal in real time camera feed. The overall methodology of the deployment and application of the combined pipeline on camera is given bellow:

Object Detecion using YOLOv11: The system takes video frame from webcam and detect persons using yolo11 model in in real time. Bounding boxes are used to locate the identified people in YOLOv11.

Background person identification: Given the persons in the frame detected, using foreground and background person separation algorithm the foreground and background person will be identified. The foreground person will remain as usual and the background persons will be processed for inpainting in further.

Mask Generation: A binary mask will be created covering the bounding boxes of all background persons except the closest one. This mask indicates the regions in the frame where persons should be removed or replaced.

Inpainting with LaMa: The mask and original frame will be fed into the LaMa inpainting model which will fill the masked regions by creating realistic background textures. LaMa When the background people are masked out, background pixels are is restored

Real-Time Processing and Display: The overall detection-mask-inpainting pipeline will run in real-time on frames from the webcam, showing real-time video, where background humans are automatically removed and their missing regions are realistically filled.

This approach uses both object detection to accurately localize the people and advanced image inpainting to convincingly remove and replace the background in real time. It supports automatic in-camera background person removal without any manual process.

3.10 Summary of the Methodology

The methodology provides comprehensive framework for the overall procedure of background object detection and removal from the image. The frame work combines working principle of the detection model and the robust inpainting model to ensure coherency of each image. This ensures excellence in properly detecting any background object and inpaint it with the background to make the object totally disappear.

Later the second part of the methodology involves deployment of the whole combined framework in a camera for making the whole procedure operate in real time. For perfect

real time experience of the whole procedure a specific workflow is provided by author that includes inclusion of proper library of accessing camera and optimization of the performance of the combined model on each frame. For perfect real time experience of the whole procedure a specific workflow is provided by the author that includes inclusion of proper library of accessing camera and optimization of the performance of the combined model on each frame.

CHAPTER IV

Implementation, Results and Discussions

4.1 Introduction

This chapter details the overall implementation process of the proposed system including the experimental setup, dataset collection and preparation process, different implementation steps and result analysis of each implementation process and evaluation metrics for overall comparison of each step of the implementation. The real time performance was evaluated with FPS values calculation per frame and overall speed calculation.

4.2 Experimental Setup and Dataset Description

The environment setup was done by the author in Kaggle environment. The T4 GPU was utilized for the training, inference and evaluation of each of the framework during the implementation process. The necessary tools used for the implementation process includes Pytorch and OpenCV.PyTorch is an open-source deep learning framework that is used to develop, train, and deploy neural networks. It's flexible and allows reduces the complexity of exchanging signals between deep learning models and the libraries they abstract over compared to some other deep learning libraries.Pytorch implements deep learning models including CNNs (image classification),YOLOv11 (object detection), and SAM (segmentation model). Pytorch includes modules and functionalities necessary for YOLOv11 object detection, SAM models mask generation and LaMa inpainting procedure.

OpenCV is another library tool that is used for the efficient preprocessing and visualization that complements the PyTorch. The synergy between these two library enabled the author building robust, flexible and real time computer vision system. The dataset collected for the implementation process is the custom dataset with single or multiple person on different complex background.

The dataset includes scenes with complex backgrounds, diverse objects, and occlusions to simulate real-world inpainting scenarios. The dataset was first manually labeled in the roboflow for proper annotation before the finetuning. Different augmentation techniques such as rotation, flipping, and scaling, enhancing the model's robustness to different input variations was performed on the dataset to ensure consistency of each image. The dataset is also divided into three parts which are train, test and valid. Table 4.1 describes the overall dataset description:

Table 4.1: Dataset description

Dataset Name	Description	Image size	No. of images
Person	random person images with label	640×640 pixels	771

4.3 Evaluation Metrics

For evaluating the whole system, both quantitative and qualitative evaluation are needed. This subsection describes the evaluation metrics used in this thesis work. As there are basically two portions of the major workflow i.e. (a) Detection and (b)Recognition. For Detection, this research primarily relied on Recall, Precision and F1-Score. For Recognition task some metrics are used in this thesis for evaluation, for example, Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM) etc.

4.3.1 Peak Signal-to-Noise Ratio

The Peak Signal-to-Noise Ratio (PSNR) represents the relationship between the maximum power of a signal and the power of its noise. To compute the PSNR, the Mean Squared Error (MSE) should be determined initially between the generated image (G) and the target image (T). The MSE is obtained using equation 4.1, where w × h represents the image dimensions, and c denotes the number of channels. Subsequently, the PSNR for the generated image is calculated using equation 4.2. The maximum possible pixel value is 255 for an 8-bit RGB image. A higher PSNR value signifies superior image generation.

$$MSE(T, G) = \frac{1}{w \times h \times c} \sum_{i=1}^{w} \sum_{j=1}^{h} \sum_{k=1}^{c} [T(i, j, k) - G(i, j, k)]^2 \tag{4.1}$$

$$PSNR(T, G) = 20 \log_{10} \left(\frac{255}{MSE(T, G)} \right) \tag{4.2}$$

4.3.2 Structural Similarity Index Measure

The Structural Similarity Index Measure (SSIM) is used to measure the similarity between two images by comparing three key features from an image: luminance *l*, contrast *c*, and structure *s*. It combines these three metrics to measure the images. These metrics are then merged to assess image similarity using Equation (**??**). In this equation, c_1, c_2, and c_3 serve as regularization constants, ensuring stability during division, especially for images with mean or standard deviation values close to zero. The σ_{TG} is the covariance of the target image T and generated image G. μ_G and σ_G^2 are the mean and variance values of image G.

A higher SSIM value reflects superior results in the comparison of images.

$$SSIM(T, G) = l(T, G)\ c(T, G)\ s(T, G) = \frac{2\mu_T\mu_G + c1 \quad 2\sigma_T\sigma_G + c2 \quad \sigma_{TG} + c3}{\mu_T^2 + \mu_G^2 + c1\ \sigma_T^2 + \sigma_G^2 + c2\ \sigma_T\sigma_G + c3} \tag{4.3}$$

4.3.3 Recall, Precision and F1-Score

Recall, Precision, and F1-Score are used to evaluate classification performance by comparing true positives (TP), false positives (FP), and false negatives (FN). These metrics are combined to assess model accuracy using equation 4.4. In this equation, TP represents correctly predicted positive cases, FP represents incorrectly predicted positive cases, and FN represents missed positive cases. Recall measures the fraction of actual positives correctly identified and it is obtained using the equation 4.4. Precision measures the accuracy of positive predictions and is obtained using the equation 4.5 and F1-Score is the harmonic mean of Recall and Precision, balancing the two metrics and calculated using equation 4.6 . Higher values indicate better classification performance.

$$\text{Recall} = \frac{TP}{TP + FN} \tag{4.4}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{4.5}$$

$$F1\text{–score} = 2 \times \frac{Tp \times Fp}{TP + FP} \tag{4.6}$$

4.3.4 Frames Per Second (FPS)

Frames Per Second (FPS) is used to evaluate the real-time performance of the proposed pipeline. It indicates how many frames the system can process and display every second, which directly reflects the responsiveness and practical usability of the pipeline for live camera-based operation. The FPS is calculated using equation 4.6, where N represents the number of frames processed within a given time interval, and t_{end} and t_{start} represent the timestamps recorded at the end and the beginning of that interval, respectively.

$$FPS = \frac{N}{t_{end} - t_{start}} \tag{4.6}$$

In this implementation, the FPS was recalculated and overlaid on the output frame after every 10 processed frames. The frame counter N was incremented for each frame passed through the detection-mask-inpainting pipeline, and the elapsed time was measured using the system clock. Once 10 frames were processed, the FPS was computed using equation 4.6 and displayed on the live output window, giving the user continuous feedback on the real-time

processing speed of the system.

To assess overall performance across the full runtime, the Average FPS was also computed using equation 4.7, where N_{total} is the total number of frames processed during the entire session and T_{total} is the total elapsed time of the session.

$$FPS_{avg} = \frac{N_{total}}{T_{total}} \tag{4.7}$$

A higher FPS value indicates smoother, more responsive real-time performance, whereas a lower FPS suggests processing bottlenecks, often introduced by the inpainting stage of the pipeline.

4.4 Implementation

According to proposed methodology in chapter III the image must go through several stages. The stages are described bellow:

4.4.1 Fine-Tuning of YOLOv11 model

A pretrained YOLOv11 model was used for the proposed method. The YOLOv11 model was fine tuned with the custom dataset images. The total dataset images were labeled with roboflow to make annotation file which represents the confidence score of each bounding boxes in images. Among the 771 images, a 80-20 split was done to define train and validation set. Kaggle (uses python 3) was the platorm used for this operation. Essential packages/libraries of python that was used are: Pytorch and Ultralytics. The trained YOLOv11 model was evaluated using standard metrics that included precision, recall, and mean Average Precision (mAP) to quantify detection precision. The training parameters are given below in the table:

Table 4.2: YOLO Training Parameters

Parameter name	**Value**
Epochs	30
Imgsz	640
batch	8

4.4.2 Testing of Trained YOLO Model

The author provides test image n as input to the trained YOLOv11 model to check its overall performance. The YOLO module is first imported from the ultralytics library. The inference step includes collecting the test images and feeding them into the trained model. The model processes the input and generates bounding boxes along with class probabilities for detected

objects. In addition, the confidence scores for each detection are analyzed to measure the reliability of the predictions. Non-Maximum Suppression (NMS) is applied to remove overlapping bounding boxes, ensuring that only the most accurate detections are retained. The outputs are then compared against the ground truth annotations to evaluate the precision, recall, and overall accuracy of the model. The model was evaluated with the detection value of the YOLOv11 model that calculates the overall method and procedure. This testing phase not only validates the effectiveness of training but also highlights potential limitations such as missed detections or false positives. The figure 4.5 bellow shows the test result:

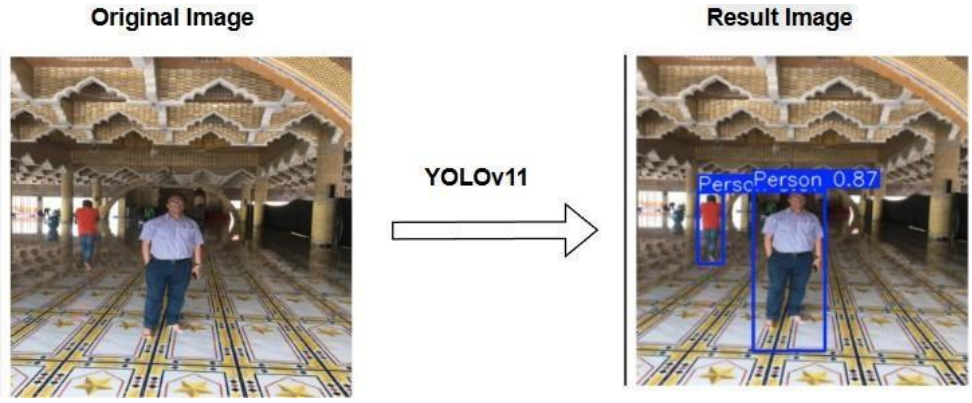


Figure 4.6: YOLOv11 Object detection

From the figure it can be said that the YOLOv11 successfully captures and calculates the confidence score of the detected person. The confidence score calculated for the foreground person is 0.90. The confidence score calculated for the foreground person is 0.90. This high confidence score indicates that the model is confident in its detection. The bounding boxes accurately cover the people in the image, with minimal overlap or missed regions. This high confidence score indicates that the model is confident in its detection. The bounding boxes accurately cover the people in the image, with minimal overlap or missed regions. The model maintains high precision and recall, indicating not only that it correctly detects most true positive instances but also minimizes false positives. This balance is critical for reliable detection in real-world applications where both missed detections and incorrect identifications can have significant consequences. Also the consistency in bounding box placement suggests that the model has been trained effectively to generalize across varied backgrounds and lighting conditions. The clear separation of multiple individuals in the frame highlights its robustness in handling crowded or complex scenes. Such reliable detections are particularly useful for downstream applications like tracking, behavioral analysis, and activity recognition.

4.4.3 Mask Image Generation

The outcome image of the Fine-tuned YOLOv11 model is then passed through the paradigm of mask generation. In the paradigm first from the outcome image the background and foreground persons are separated using either algorithm approach 1 or 2. Then the images are fed through the trained SAM ViT model. The background persons are segmented using pretrained SAM ViT model. The SAM model in instance segments the background persons in the image frame. The OpenCV library finds the segmented part of the image. The python OpenCV takes the result from SAM and then it converts it to numpy values which are set to 255 for 1 and 0 for 0 values. The final mask is a binary image with 255(White) for the person to be removed and and 0(Black) for everything else to be preserved. This binary mask serves as the crucial intermediate representation that guides the subsequent inpainting stage. Once generated, the mask ensures that only the designated background persons are targeted for removal while preserving the integrity of the main subject and the surrounding context. The precision of this mask directly impacts the quality of the inpainted output, making accurate segmentation a critical step in the pipeline. . The mask image generation can be described through the figure bellow:

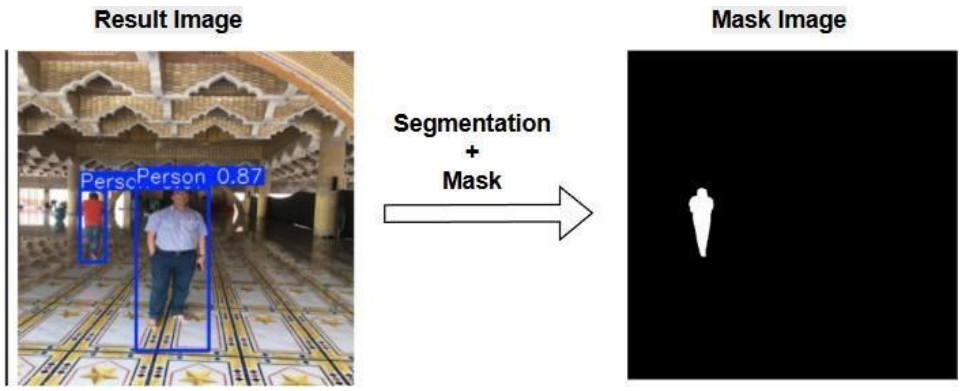


Figure 4.7: Mask Generation

4.4.4 Testing of LaMa model

Later after the end of training the inference of the LaMa model was implemented by the author. A random image with mask was sent to the trained model as input to visualize the performance of the trained model. The figure 4.3 shows the overall result.

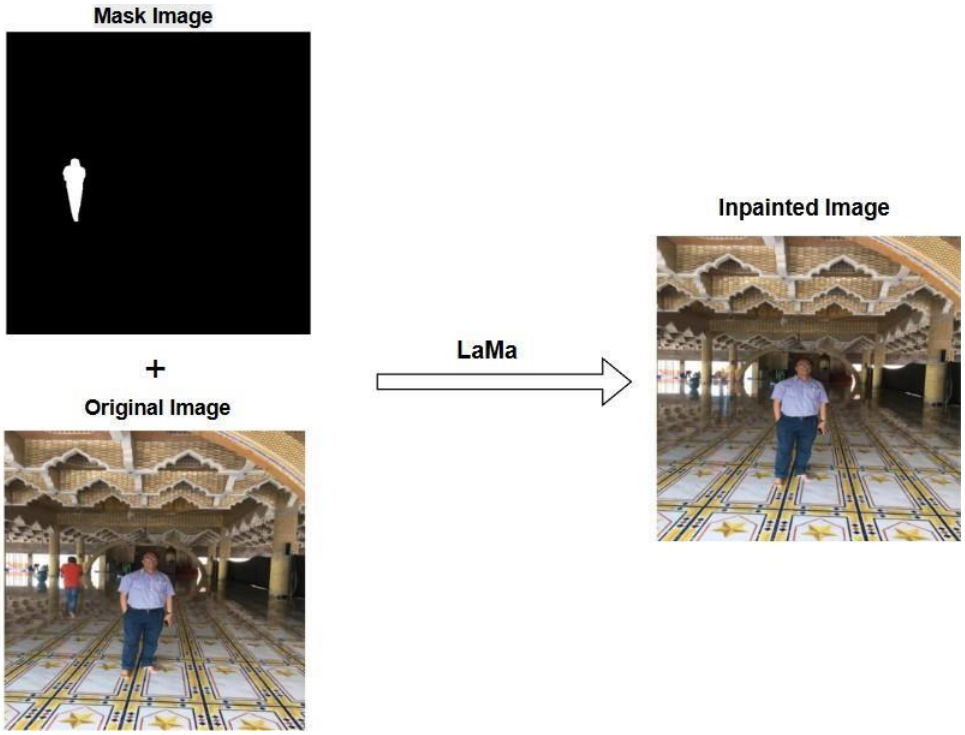


Figure 4.8: Inpainting with LaMa

4.4.5 Deployment of Overall Process in Camera

The final step of implementation was to enable the whole pipeline performs in real time. After the implementation and performance analysis of the each framework of the pipeline the whole pipeline was deployed in a camera for real time operation. The overall description of each method of the deployment procedure is given bellow:

Initialization of libraries: At first, the necessary libraries to access the camera was imported. These libraries include openCV libraries to access the camera. The other library includes pytorch and YOLO. These libraries provide necessary ways to operate on each camera frame.

Loading Models: Two models are loaded to operate on the camera frame to perform real time object removal and inpainting. The two models are YOLOv11 paired with SAM model and LaMa model. The YOLOv11 model is loaded to detect all persons in-front of Web Camera in real time with SAM ViT based model to segment and mask only the background persons properly. Then the LaMa model is loaded to reconstruct the mask part with background.

Opening webcam and accessing Camera Frame: The webcam is opened with the openCV library. The webcam is opened with proper resolution settings.

Capturing frame from webcam: After accessing the webcam it catches each frame. Each camera frame is treated as image with persons and background. The live stream camera feed is then processed on each step.

Automatic Mask Generation: The Mask generation procedure follows the algorithm derived to separate foreground person and background person in real time. The person closest to the camera remain unmasked and the background persons are masked and the overall process done automatically using YoLov11 model detection.

Checking Mask image for mask region: The decision is then made to check for masked region on mask frame before feeding to LaMa each frame. If the masked region is found then it is fed to LaMa model to operate else frame is skipped and original frame is shown as output on display.

Running LaMa inpainting on Masked region: The FFCbased LaMa model performs the inpainting on the masked region. It frills the masked region with the background which implies the removal of the background person with the background.

Calculation and Overlaying FPS every 10 frames: The processing speed and display frames per second on the output was kept tracking for the feedback of the user of the system. FPS provide indications of how many frames the program can process and output every second, which is a key measure of real-time performance and responsiveness. Each passed frame to the pipeline is counted in this process.

Displaying the output frame to the user: The resultant inpainted output frame is then shown live to the user in a window. **Checking if the user requested to quit by pressing 'q':** A functionality to check whether the user wanted to keep the webcam open or closed was also added for terminating the operation.



Figure 4.9: Captured Normal Camera Frame



Figure 4.10: Captured Camera Frame after deployment of the system

4.4.6 Camera Performance Optimization

After the camera deployment the overall process seemed little slow since the trained model was heavy. The author had to optimize the YOLO side first. The optimization process involved running the YOLO every N(Can be 5) frames and then using the lightweight tracker in between. LaMa side optimization involved resizing each camera frame before inpainting it. After the inpainting process is done then again resizing the whole camera frame back to the previous resolution. The overall pipeline speedups then involved processing 2-3 frames at once when the GPU had good memory. After the camera deployment, the overall process seemed a little slow since the trained model was heavy. The author had to optimize the YOLO side first. The optimization process involved running the YOLO model every N (can be 5) frames and then using a lightweight tracker in between these frames to reduce computation and speed up processing. This approach balances accuracy and efficiency by limiting the number of expensive detections. On the LaMa side, optimization involved resizing each camera frame to a smaller resolution before inpainting it, significantly reducing the computational load. After the inpainting process was done, the whole camera frame was resized back to its original resolution to maintain visual quality. Additionally, the overall pipeline speedup was achieved by processing 2-3 frames simultaneously when the GPU had sufficient memory,

4.5 Result

Result section presents both quantitative and qualitative results . quantitative results usually consist of numerical data and statistical analysis. They are often illustrated with tables, graphs or charts and regarded as objective results. On the other hand, qualitative results include images, narrative, pattern and as regarded as subjective result. Together, these two

approaches offer a comprehensive understanding of the research outcomes.

4.5.1 Quantitative Result of Detection Phase

The proposed model was trained on a custom dataset of random people. The evaluation process involved calculating the precision, recall, and F1 score of the overall model to quantitatively assess its performance in accurately segmenting people in various scenes. Precision measured the proportion of correctly predicted person pixels out of all pixels classified as person by the model, while recall quantified the proportion of true person pixels successfully detected among all actual person pixels. The F1 score was computed as the harmonic mean of precision and recall to provide a balanced measure accounting for both false positives and false negatives. These metrics collectively enabled a comprehensive understanding of the model's effectiveness, robustness, and generalization ability on unseen images. Table 4.3 provides the evaluation metrics of the proposed model:

Table 4.3: Evaluation Metrics Results

Evaluation Metrics	Values
Precision (P)	0.801
Recall (R)	0.806
F1 Score	0.831

The table 4.4 provides comparative analysis of the different detection method.

Table 4.4: Comparison of proposed model with different detection method

Models proposed by	Image Number	mAP(%)	F1-Score
C.Kim *et al* [2]	125914	95.62%	0.92
M. Sanchez *et al* [13]	630	80.1%	0.886
S.Jain *et al* [6]	690	98.5%	0.81
Proposed Methodology	771	89.37%	0.83

4.5.2 Qualitative Evaluation of Inpainting Result

LaMa model was utilized in the system to perform the person removal inpainting properly for any complex environment. The overall result of the outcome image after inference of the LaMa model has completely provided better result. LaMa's advanced inpainting technique results in superior inpainting quality and efficiency for the images. It maintained the visual coherence and consistency of inpainting the missing part of the image. The ghosting effect and artifacts were reduced as much as possible. The LaMa model's robustness in handling large missing regions and intricate details makes it well-suited for diverse, real-world images

with complex backgrounds. Its architecture, which leverages resolution-robust Fourier convolutions, enables high-quality inpainting even at high resolutions without excessive computational overhead. This results in natural, visually appealing outcomes where the inpainted areas blend seamlessly with the rest of the image, preserving textures and geometric patterns. The model also significantly reduces common issues like blurring, ghosting, and edge artifacts, which enhances the overall user experience. These strengths make LaMa a powerful tool for person removal tasks, enabling effective background completion and content-aware editing in varied and challenging environments. The LaMa model is also highly flexible and adaptable to various input sizes and mask shapes, making it applicable to a wide range of image editing scenarios. Its ability to maintain structural and textural coherence even with large missing regions sets it apart from many traditional inpainting approaches. Moreover, its optimized architecture allows for efficient inference, enabling near real-time processing which is critical for practical applications. By leveraging resolution-robust Fourier convolutions, the model generalizes well beyond its training resolution, preserving fine details and minimizing blurriness.Furthermore, LaMa effectively reconstructs contextual information by learning the spatial and semantic relationships surrounding the masked regions. This capability allows the model to generate plausible background structures even when a significant portion of the image has been removed. In person removal applications, this is particularly beneficial because the removed regions often contain complex textures, objects, and background patterns. LaMa can preserve important visual characteristics such as edges, colors, textures, and geometric structures during the reconstruction process. The generated content therefore appears more consistent with the surrounding areas, reducing noticeable transitions between the original and restored regions. Its ability to process irregular and diverse mask shapes also makes it suitable for removing multiple objects or people from a single image. In addition, the model demonstrates good generalization when applied to images containing different lighting conditions, environments, and background compositions. The combination of reconstruction quality and computational efficiency makes LaMa suitable for integration into automated image editing and real-time object removal systems. The figure 4.10 gives visual understanding of the output result images.

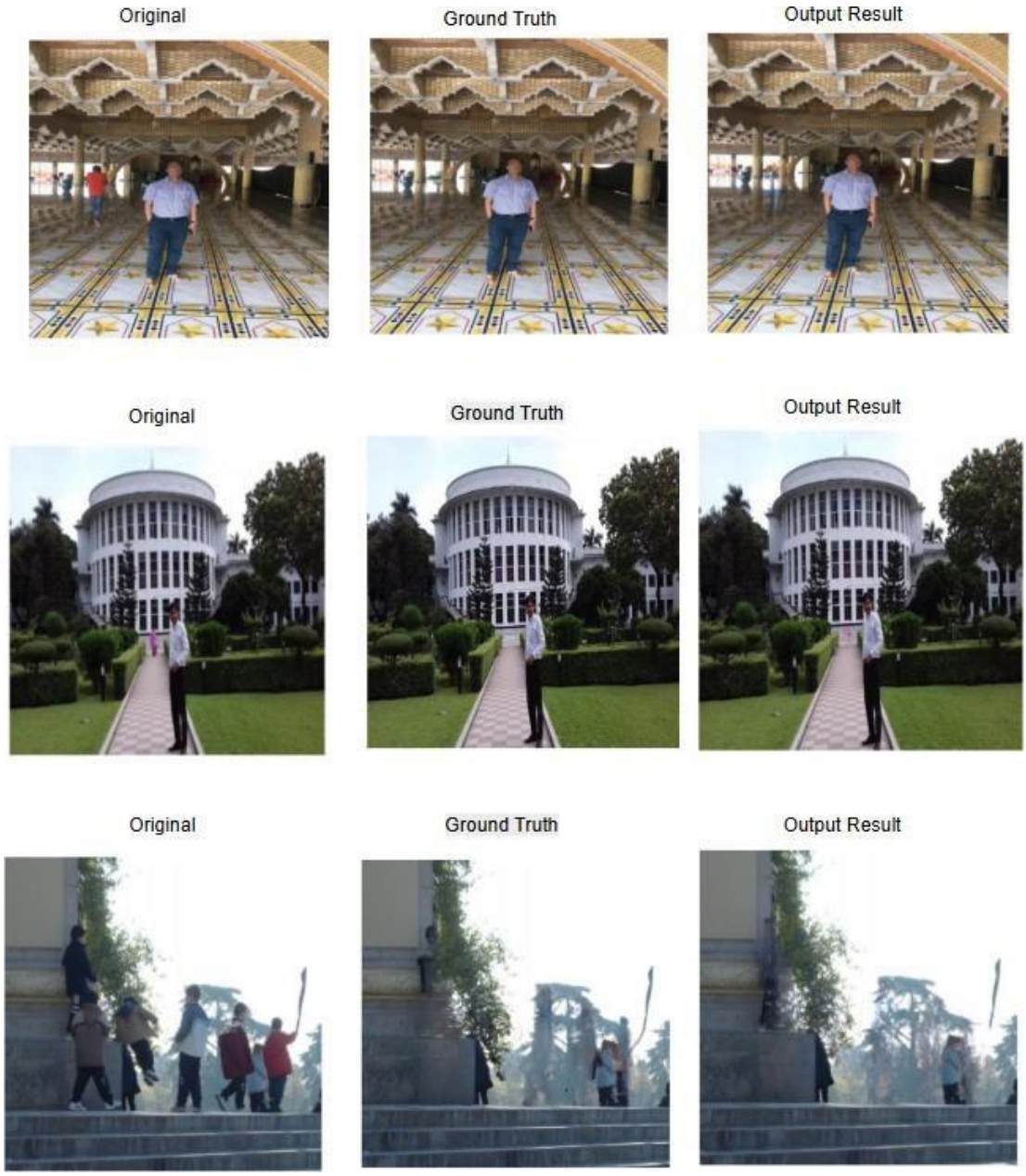


Figure 4.11: Model Results in proposed pipeline

4.5.3 Quantitative Evaluation of Inpainting Result

For the quantitative evaluation of the inpainting result, evaluation matrices like the PSNR and SSI were calculated. The SSI value calculated was 0.9022 which is quite impressive value that indicates quite better structural and perceptual similarity between the original and inpainted images. For the quantitative evaluation of the inpainting result, evaluation matrices like PSNR and SSIM were calculated. PSNR (Peak Signal-to-Noise Ratio) and SSIM (Structural Similarity Index) are two common evaluation metrics utilized to measure the quality of the reconstructed images compared to the reference (original) image. PSNR

measures the ratio between the maximum possible power of an image converted to a signal and the power of corrupting noise that affects the overall fidelity of the image's representation. It is measured in decibels (dB). The PSNR value achieved for the model is 24.89 dB, which is moderate for image inpainting quality.

Table 4.5: Comparison of proposed model with different inpainting method

Models proposed by	**PSNR(dB)**	**SSI**
C.Kim *et al* [2]	27.62%	0.92
M. Sanchez *et al* [13]	28.10%	0.886
S.Jain *et al* [6]	24.55%	0.81
Proposed Methodology	24.89%	0.90

4.5.4 FPS Evaluation of the Proposed Model

The real-time performance of the proposed pipeline was evaluated by measuring the per-stage processing time of each component during continuous execution. The total processing time per frame was calculated by summing the average processing time of the individual stages, as shown in Equation 4.8.

$$T_{frame} = \frac{1}{k}, T_{YOLO} + T_{mask} + T_{LaMa} + T_{overhead} \tag{4.8}$$

where T_{YOLO}, T_{mask}, T_{LaMa}, and $T_{overhead}$ represent the measured average processing times (in milliseconds) of the detection, mask-generation, inpainting, and frame capture/display stages, respectively. The parameter k represents the frame interval at which the YOLO11 detection model is executed. In the proposed implementation, $k = 3$, meaning that the detection results are cached and reused for the following two frames to reduce the computational overhead of repeated detection.

The frames-per-second (FPS) performance was calculated from the measured total processing time using Equation 4.9.

$$FPS = \frac{1000}{T_{frame}\ (\text{ms})} \tag{4.9}$$

Table 4.6 presents the measured processing time of each stage of the proposed pipeline. These measurements were obtained during the actual execution of the system on the GTX-class GPU used for development. Unlike theoretical estimates, the reported values reflect the practical computational cost of the complete pipeline, including detection, mask generation, LaMa-based inpainting, and frame capture/display.

Based on the measured total processing time of approximately 315 ms per frame, the proposed pipeline achieved an FPS of approximately **3.2 FPS**. This result was obtained from the actual

Table 4.6: Estimated per-stage processing time of the proposed pipeline

Stage	Estimated Time (ms)	Basis
YOLO11 Detection	20–30	Ultralytics benchmark, scaled for non-optimized GPU
Mask Generation	1–3	CPU-based NumPy/OpenCV operation
LaMa Inpainting	250–350	Estimated from FFC network complexity
Frame Overhead	5	Webcam capture and display
Estimated Total	**≈315**	

continuous execution of the complete system rather than from published benchmark values or theoretical complexity estimates.

The LaMa inpainting stage was identified as the primary computational bottleneck because it required substantially more processing time than the detection and mask-generation stages. Despite this computational cost, LaMa was retained as the primary inpainting model because high-quality reconstruction of large missing regions was a central objective of the proposed person-removal system. The results demonstrate a trade-off between computational efficiency and background reconstruction quality, where the use of LaMa contributes to visually coherent and natural-looking inpainting while limiting the overall processing speed of the pipeline. The measured FPS demonstrates the practical real-time processing capability of the proposed person removal system on the development hardware. However, the achieved frame rate depends on factors such as input resolution, mask size, number of detected persons, GPU utilization, and background complexity. Therefore, the reported **3.2 FPS** represents the measured performance under the experimental configuration used in this study.

Table 4.7: Comparison of LaMa with other image inpainting approaches

Model	Approach	Large-Mask	Resolution	E*A*ciency
LaMa	Fourier Convolution	Strong	Strong	High
DeepFill v2	Contextual Attention	Good	Moderate	High
EdgeConnect	Edge Prediction	Moderate	Moderate	High
RegionWise	Region-Based	Good	Moderate	Moderate
AOT-GAN	Aggregated Transformations	Good	Good	High
CoModGAN	Conditional GAN	Strong	Good	Moderate
MADF	Multi-Scale Features	Strong	Good	Moderate
Proposed System	YOLO11 + SAM/SAM2 + LaMa	**Strong**	**Good**	**3.2 FPS**

4.6 Achieved Objectives

The main objective of the author is to implement a pipeline that performs the background object detection and removal well on the real time camera. This objective is achieved after the overall implementation process. The other side objectives were utilizing perfectly an efficient person detection model and inpainting model on real time camera and combining them to form a hybrid pipeline. This objective was also achieved.

CHAPTER V

Societal, Health, Environment, Safety, Ethical, Legal and Cultural Issues

5.1 Ethical Consideration of the Work

The author acknowledges that the project raises serious ethical concerns. The faces of people who have been recorded can be manipulated without their permission. The dataset collected was custom dataset of authors and different person was used in dataset. There was approval of persons. Detection models could also be biased, potentially resulting in asymmetric treatment of different demographic groups. Further, if not appropriately managed, such technology can raise surveillance and misuse threats for personal data. To combat these issues, transparency, informed consent, and secure data stewardship are crucial. It is also important to establish fairness and accountability in model design and deployment. These ethical issues need to be addressed in this field of research.

Moreover, there is a risk that such technologies could be employed for mass surveillance without adequate legal or societal oversight. Ensuring data anonymization and minimizing personally identifiable information is critical to protecting privacy. The potential for deepfake generation increases the need for responsible use guidelines and regulatory frameworks. Researchers should also consider implementing bias audits and regular monitoring to detect unintended discriminatory behavior. Engaging with ethicists, policymakers, and affected communities can help guide responsible development. Public awareness and education about the limitations and risks of such systems are equally important. Finally, fostering a culture of ethical AI development within the research community can help mitigate potential harms and ensure that technological advancements benefit society as a whole. In addition, international cooperation may be necessary to establish globally accepted standards for ethical AI use, as misuse in one region could have cross-border impacts.

Developers should also prioritize transparency in explaining how models make decisions, so that stakeholders can better understand their strengths and weaknesses. Independent auditing bodies could play a role in ensuring compliance with ethical principles and legal requirements. Strong data governance policies must be enforced throughout the system lifecycle, from collection to deployment. Researchers must also recognize the psychological and societal impacts of surveillance technologies, which may lead to public distrust if left unchecked. Balancing innovation with civil liberties requires ongoing dialogue between

technologists, regulators, and society. By promoting responsible design and establishing clear lines of accountability, the risks associated with misuse can be substantially reduced. Ethical foresight should therefore remain at the center of future research, ensuring that advances in AI contribute positively to human welfare and social progress.

5.2 Impact of the Work on Societal Issues

The implemented method can create a significant impact on society. It may even facilitate privacy by providing a way to keep individuals in video feeds anonymous or remove them entirely, which is especially useful in public surveillance scenarios or high-security areas. At the same time, it could raise concerns regarding enhanced surveillance capabilities and the potential misuse for unauthorized monitoring. The technology could help bolster security and safety by focusing on signs and objects of interest without revealing private citizen identities. However, it may also introduce biases or exclusion if detection models are not designed inclusively and fail to account for diverse populations. Moreover, the adoption of such systems demands strict regulatory frameworks to ensure ethical deployment. Continuous monitoring, auditing, and validation of model performance are essential to prevent unintended harms. Public transparency about how the technology is used and what data is collected is crucial for building societal trust. Training datasets should be diverse and representative to mitigate bias and ensure equitable outcomes. Collaboration with ethicist, policymakers, and affected communities can guide responsible implementation.

Awareness campaigns and educational initiatives can help the public understand the benefits and limitations of these technologies. Ultimately, while the method provides enhanced privacy, security, and safety, it must be carefully controlled and governed to prevent ethical and social misuse. Balancing technological advancement with societal norms is key to gaining trust and acceptance in AI-driven surveillance applications. Moreover, the deployment of such systems may shape public discourse around the acceptable boundaries of surveillance in democratic societies. It may also encourage governments to update existing data protection laws to accommodate AI-specific challenges.

In industry, adopting privacy-preserving detection frameworks could improve consumer trust and brand reputation. At the same time, these approaches could set competitive standards that push other companies toward more responsible innovation. The societal implications further extend to journalism and media, where anonymization can protect vulnerable groups such as protestors or whistleblowers. If carefully regulated, the technology could provide a blueprint for international cooperation in AI ethics and governance.

Finally, the continued integration of social values into technical design ensures that innovation advances without compromising fundamental human rights. Limited access to identifiable video data, such as managed permission systems or temporary video access, also contributes

to trust. It is designed so that, personnel of concern, such as installation engineers or security officers can complete required tasks without having permanent or unrestricted access to sensitive material giving them less opportunity to abuse privileges.
This responsible data stewardship, alongside transparency and user control, can help convince the public that surveillance can be in everyone's interests, rather than seen as a snooping threat. Minimizing the risk of consent problems and incursions of privacy prevent potential harm to mental health related to the sense of being under surveillance and intruded upon.

5.3 Impact of the Work on Health Issues

The effects of the work is in general indirect but also significant. By offering increased privacy to prevent individuals from appearing within camera frames, it could help to alleviate feelings of stress and anxiety about always being watched. Misuse or lack of consent have mental health implications including feelings of privacy violation. It also facilitates health-related applications such as social distancing monitoring or health condition detection based on computer vision.
The effectiveness and accuracy of the technology are qualities that can be used to improve and ensure safer environments, but the ethical use should be implements to avoid negative psychological conditions associated with privacy invasion. In general, if responsibly used, it can be a boon to public health. Additionally, the proposed system can contribute to fostering trust in AI-driven surveillance when individuals feel their identities are adequately safeguarded.
By reducing the exposure of personal identifiers, it also lowers the risks of identity theft or unauthorized profiling. Such measures are particularly relevant in educational institutions, workplaces, and public transport systems where surveillance is increasingly being adopted. On a broader scale, this work may influence the formulation of new privacy-preserving policies and guidelines for smart city infrastructures. It can also support research into human-centered AI by providing a model for balancing safety with individual freedoms. Importantly, if integrated into healthcare facilities, the technology could enhance patient monitoring without infringing on dignity and autonomy.
Beyond health, the approach might extend to areas such as crowd management in large gatherings, ensuring safety while maintaining anonymity. Furthermore, the demonstration of responsible design in this project can serve as a reference for future innovations that aim to harmonize technology with social values. Ultimately, the societal effect lies not only in technical contributions but also in setting a precedent for ethically sensitive AI development. risks of such systems are equally important. Finally, fostering a culture of ethical AI development within the research community can help mitigate potential harms and ensure that technological advancements benefit society as a whole.

In addition, international cooperation may be necessary to establish globally accepted standards for ethical AI use, as misuse in one region could have cross-border impacts. Developers should also prioritize transparency in explaining how models make decisions, so that stakeholders can better understand their strengths and weaknesses.

Independent auditing bodies could play a role in ensuring compliance with ethical principles and legal requirements. Strong data governance policies must be enforced throughout the system lifecycle, from collection to deployment. Researchers must also recognize the psychological and societal impacts of surveillance technologies, which may lead to public distrust if left unchecked. Balancing innovation with civil liberties requires ongoing dialogue between technologists, regulators, and society. By promoting responsible design and establishing clear lines of accountability, the risks associated with misuse can be substantially reduced. Ethical foresight should therefore remain at the center of future research, ensuring that advances in AI contribute positively to human welfare and social progress. In addition, there is a need to embed fairness and non-discrimination principles into the design of AI systems in order to prevent reinforcement of bias and to guarantee equitable treatment to all people. The more transparent models are in their operations and decisions, the more trustworthy they become and the more accountable they can be held. Continued human management is necessary, especially in cases where AI influences our most essential realms of life. Development should be guided by ethical governance frameworks that support society's flourishing and protect civil liberties. There may be a need for international cooperation to develop norms that prevent abuse and protect privacy at the global level. Akanthou More inclusive discussion between technologists, policy-makers and communities will facilitate a balancing of innovation with human rights. In the end, any ethical AI surveillance should be at least as concerned with the respect, dignity, and autonomy of individuals, as with their security and efficiency.

5.4 Impact of the Work on Safety Issues

Utilizing the YOLO v11+LaMa pipeline for real-time automatic background person removal has far-reaching safety implications, on both the good and the bad side. On the one hand, it is believed that this technology can improve the safety and personal security of an individual by being able to provide privacy protection in both public and private environments. It minimizes risks of unauthorized ID or monitoring (stalking, harassment, data theft) by cutting out background people from the images (or video stream). This is true especially in situation such as virtual meetings or sharing on a social network, where users would want to exercise control over their digital persona. But there are also safety issues we need to think about." The real-time manipulation of camera images increases the danger of loss of important visual environmental information.

Background people can be cues for situational awareness — like if there are any threats, or if an emergency occurs; automatically deleting them could inadvertently obscure safety risks or nefarious activities from the viewer, or any monitor systems. For safety critical applications in public surveillance, traffic monitoring or workplace security, this may involve a delay in recognizing risky situations or accidents. Furthermore, another important issue in this technology is the accuracy of the detector.

Any missed identifications of people in the YOLOv11 model, or mask generation errors, could result in either the accidental removal of critical individuals or residual artifacts in outputs that mislead interpretation. False negatives might mean overlooking persons who are of interest for security assessments, while false positives might remove people incorrectly and therefore interfere with trusted monitoring. There are also the problems of real-time interfaces between the LaMa inpaintings and these detections. While LaMa provides visually convincing backgrounds for the masked region, observers might be misled regarding actual scene contents by the synthetic reconstruction, which raises trust and accountability issues. Ultimately, the application of this technology also needs to be ethically and legally managed to ensure that it is used in a responsible manner that does not support harmful deception and complies with public safety mandates. Safety standards and policy should recommend where and when real-time background person removal is appropriate, weighing privacy advantages against possible threats to public and personal safety.

Overall, while this project may have the potential to support privacy and data protection as a safety benefit; it also poses risks of decreasing situation awareness, false detections, and trust. Mindful roll out along with proper testing is going to be crucial to avoid any safety concerns along with better privacy enhancements. In addition, there is a concern about the possibility of privacy infringement caused by mismanagement or unauthorized access to real-time video streams and image data which can potentially be used by malicious sources to engage in identity theft and stalking. It could create unintentional deepfake or manipulated video, leading to ethical implications of deception and trust that could have effects on safety protocols.

And reliance on the AI models themselves exposes the system to the dangers of faults, whether due to defective hardware, software or of programming errors that could result in important monitors going blind to evolving hazards. He stressed the importance of having human presence on site and fail-safe systems in deployments with safety-critical functions. Moreover, as the reliance on automated systems grows, there is a danger of over-trust in the technology, where operators may become complacent and fail to cross-check outputs with human judgment. This could lower vigilance and amplify risks in dynamic environments. Continuous retraining and validation of the YOLOv11 + LaMa pipeline are therefore necessary to maintain performance in diverse and evolving real-world scenarios. Another point of concern is adversarial attacks, where maliciously crafted inputs could bypass

detection or force incorrect inpainting results, thereby undermining both privacy and safety. Regular auditing, transparency in system design, and accountability frameworks should be integrated into deployments to ensure ethical use. Finally, proper user training and awareness programs will be essential so that individuals and organizations understand both the benefits and limitations of the technology before relying on it for safety-critical decisions.

5.5 Impact of the Work on Legal Issues

The legal consequences of this kind of project were legally effected with RT-ABPR of persons is related to privacy laws and data protection laws. Because this system is analyzing photos of people, it also has to adhere to regulations like GDPR or CCPA around consent, utilization and storage of data. Drone hackers who illegally record of modify intimate photos may be found liable. Licensing problems may also emerge in terms of the use of YOLOv11, as some releases may come with obligations related to open-source licenses such as AGPL, which impose limitations on commercial use or deployment in closed source. Responsible lawful compliance is indispensable in order to steer clear of fines and litigation: transparency, the user's permission and licenses' respect. Information security as well as the appropriate treatment of image data also are legal issues. In general, negotiating the laws of privacy, licensing, and data protection is a major legal issue in projects such as this. Besides privacy and data protection legislation, licensing concerns have a key impact on the legal framework which is to be complied when implementing solutions related to YOLOv11 and LaMa for real-time automatic background person removal.

The YOLOv11 model is often licensed under the AGPL-3.0 license, with some commercial use requirements. Under AGPL, any alterations to the software or model must release the full source code of the modified or derivative version in the open and freely available. For prescriptive licenses which want to keep proprietary code or modifications secret, this requirement can dig an enormous hole in their commercial deployment strategy. As a result, companies may also need to obtain a commercial license or enterprise license from the copyright holder(s) (Ultralytics and Roboflow*) to use and distribute YOLOv11 in a closed-source project with their proprietary YOLOv11 work even if not linked with the released YOLOv11.

Moreover, legally using such models is required to also be compatible with data protection laws such as GDPR in Europe or CC PA in California. These laws control how the personal information about individuals—images or videos of them, for example—is collected, processed, held and shared with others. Because the project would be working with, and making changes to, images of live people, it is important to get explicit user opt-in and be very clear about what is done with the data to avoid legal issues. Not following these policies can result in serious fines and legal action. Second, there is the issue of data security, since

data that derives from personal images may result in privacy breaches or misuse of information due to unauthorized access or system hacking. And the threat of drone hackers conducting unauthorized surveillance or tampering with images introduces a new legal dimension, which forces one to take cyberneticist measures extremely seriously. Finally, this project is situated at the crossroads of privacy law, intellectual property rights and software licensing requirements.

Public organizations should implement sound lawful compliance strategies, such as clear user transparency, adherence to licensing agreements by obtaining the relevant commercial rights and robust data protection controls. This complex, multi-dimensional legal compliance is essential in order to avoid risks in the face of fines, legal action or damage to reputation, to make sure that technological innovation moves forward in a secure and legal manner". Additionally, in light of technology advancements, continued legal focus and amendments to data protection legislation it is expected that the management of these types of projects will be affected by future guidance. Organizations need to keep up with new policies and legal developments around AI-based image processing. If you work for a legitimate business there should be no issues, but inherent consultant surveillance on deployment can help nip potential compliant issues in the bud. Moral factors must support legal demands to shape user trust and social acceptance.

Furthermore, transparent user agreements and documentation, in terms of how data is managed, what the purposes of the use are, the extent of its usage and model licensing, are vital. Such records not just facilitate compliance, they also contribute to the drive for greater transparency and accountability. Finally, harmonizing technology innovation with rigorous legal and ethical considerations will be essential for sustainable implementation and societal value.

Moreover, organizations should ensure that personal images are collected and processed only for clearly defined and legitimate purposes. Appropriate access-control mechanisms should be implemented to prevent unauthorized users from viewing, modifying, or distributing sensitive image data.

Data should also be securely stored and, where necessary, encrypted during both transmission and storage to reduce the risk of unauthorized access. Users should be informed about how their images are processed, whether the images are retained, and when they will be permanently deleted. Special attention should be given to images containing identifiable individuals, as improper processing may negatively affect their privacy and personal rights. The deployment of automated person removal systems should therefore consider potential misuse, including unauthorized manipulation or misleading alteration of visual content. Developers should maintain accurate documentation of the datasets, models, libraries, and third-party components used throughout the system. Open-source models and software components should be used in accordance with their respective licenses and attribution

requirements. Regular security assessments and system monitoring can further help identify vulnerabilities and reduce potential legal, ethical, and privacy-related risks.
Ultimately, responsible implementation requires a balance between technological capabilities, individual privacy, data security, legal compliance, and the broader social impact of automated image processing.

5.6 Impact of the Work on Cultural Issues

The idea of a real-time automatic background person remover with YOLOv11 and LaMa models can have strong cultural effects. In numerous cultures, the depiction of people in collective visual spaces is heavily embedded in cultural identity, norms and community values. To remove persons from these scenes the background people in particular could lead to the whites-washing of valuable social and culture-specific cues present in group situations or in public settings. Such an alteration would ribbon the world apart bit by bit, making it harder to see or to see certain cultural groups or communities, marginalizing them, possibly on purpose possibly not. Additionally, norms and practices around privacy and the use of images are very different elsewhere in the world. In a few cultures, being digitally edited or filmed without permission is a serious breach of social and personal ethics. The aromaticity and resort to scale of this technology may exacerbate consent and autonomous conditions concerns of the unsuspecting use of personal space alteration or erasure in the live environment, confronting questions of personal dignity and societal trust. There's also a cultural layer of surveillance and control that the tech essentially generates.
In cultures that were particularly sensitive to being watched and monitored, using background removal in public and/or shared spaces might raise concerns related to privacy and social control. Such fears can fuel animosity toward surveillance technologies, and can deepen systems of vulnerability and isolation for minority cultural communities. A second cultural effect concerns how it transforms social memory and historical record. Pictures and videos freeze more than mere faces; they harden cultural moments and social dynamics. The manipulation of these images consisting in deleting individuals from the scene may have been used to undermine the fictional and documentary nature of cultural and historical facts and could play a role in how the communities in question remember their past and organize their present.
Additionally, the subtle biases present in biased training sets like those for YOLOv11 can cause person detection and removal to have a greater negative effect on certain demographic or cultural groups, and continue the culture-pattern reinforcement problem. Such tech bias can bring about unfair cultural representation and brings to the fore the need for more inclusive and culturally informed AI model development. Overall, the project does deliver new technical privacy protection and user experience enhancing possibilities but has to be

taken with a deep cultural sensitivity. Clear communication, respect for a variety of cultural norms, and the integration of fire with design principles are essential if technology is to support culture, rather than undermine it. Editors Note However, recognition of these cultural dimensions is critical for the ethical and social deployment of person removal systems. Going downstream, in additional to the cultural and social impact, this project can help sway cultural narratives and norms in subtle ways.

The automatically removal of background people with live video feeds might also lead to a type of culture erasure, where the existence and importance of certain individuals, specifically marginalized or minority individuals, isn't depicted or are possibly excluded from being told as a part of visual storytelling. This distortion of visible reality reconfigures how cultural encounters and public spaces exist in documentation and memory, offering the possibility of rupturing collective cultural memory. On the whole, while applications of real-time AI image processing are cause for environmental concern, they have the potential to create efficiencies elsewhere (such as in less reliance on physical travel or manual labor in video editing) through innovative solutions. But developers and stakeholders must stay vigilant to evaluate and mitigate the ecological impacts in line with AI sustainability best practices. Transparent reporting on both energy use and emissions is also beginning to be seen as a necessity to develop AI responsibility in line with global goals for the environment.

Furthermore, automated person removal may influence how visual records are interpreted when the original context of an image is altered. Therefore, the technology should be applied carefully when images are intended for historical, cultural, educational, or journalistic documentation.

Preserving the original image alongside the processed version can help maintain the authenticity and traceability of visual information. Clear labeling of AI-processed images can also help viewers distinguish between original and modified visual content. Such transparency can reduce the possibility of users misunderstanding an altered image as an accurate representation of the original scene. The system should also avoid being used to intentionally manipulate visual records in ways that could misrepresent events or individuals.

From an environmental perspective, deep-learning models require computational resources that can contribute to energy consumption during both training and inference. The computational requirements of LaMa are particularly relevant because inpainting is performed repeatedly when processing continuous video frames. Reducing the input resolution can decrease computational requirements while still maintaining an acceptable level of visual quality.

Similarly, reusing detection results for consecutive frames can reduce unnecessary computation within the overall pipeline. The proposed implementation applies frame-level optimization by caching detection results and reducing the frequency of repeated YOLO11 inference. Such optimization can contribute to lower computational demand and improve the

overall efficiency of the system. Efficient hardware utilization is also important because unnecessary GPU processing can increase energy consumption without providing significant improvements in output quality.

Future versions of the system could investigate lightweight inpainting architectures or model optimization techniques to further reduce computational requirements. Techniques such as model quantization, pruning, and efficient inference could potentially improve energy efficiency while preserving acceptable reconstruction quality. However, any optimization should be evaluated carefully to determine whether reductions in computational cost introduce noticeable degradation in image quality.

A balanced approach is therefore required between reconstruction quality, processing speed, computational resources, and environmental impact. Documenting computational requirements can also help future researchers evaluate the sustainability of similar AI-based image-processing systems. Considering social, cultural, and environmental factors throughout the development process can support more responsible deployment of automated image-editing technologies.

Ultimately, sustainable and ethical implementation requires that the benefits of automated person removal are achieved without unnecessarily compromising cultural representation, transparency, or environmental responsibility.

5.7 Impact of the Work on Environment

The environmental implications of a system for real-time automatic background person removal with AI models such as YOLOv11 and LaMa are complex and meaningful. The major environmental issue is the huge energy consumption when training and deploying the deep learning models. When big AI models are trained, the process often demands immense computational power, paid for by energy-intensive data centers running on electricity generated by fossil fuels in a number of parts of the globe. And as they do so, they consume energy, which contributes to greenhouse gas emissions and makes climate change worse. Aside from training, the challenge of running these models non-stop for processing live video drains some of that compute. The persistent inference computation of YOLOv11 for object detection as well as LaMa for image inpainting burdens the duplex processor, resulting in prolonged electricity consumption while the system is running.

The aggregate power requirements for GPUs, CPUs and cooling in the data center or the edge contribute to the carbon footprint of the application. Furthermore, the hardware in lifecycle prior to the use (manufacturing and disposal of the GPU, and servers after use) also contributes to the environmental impacts through mining (assemblage of variety of rare minerals) and electronic waste. The water usage associated with cooling servers and data centres puts additional stress on local water supplies and has the potential to have an impact

on the local ecosystem and communities. However, progress in model optimization, energy efficient hardware, and operation of sustainable data centers may contribute to the mitigation of some of the environmental impacts. Approaches such as model pruning, quantization, and renewable energy usage help reduce AI's carbon footprint. Further, it is important to use efficient real-time computer vision pipelines that trade off computational cost for accuracy for sustainable usage. To further ellaborate on the environmental footprint of the project, the ongoing use of large models such as YOLOv11 and LaMa for real time video processing may result in a significant energy consumption, in particular when implemented on edge devices or central servers that are running 24/7.

The training of these models is computationally expensive, and requires access to energy-guzzling GPU servers and long periods of electricity use, which directly results in carbon emissions based on the energy mix of a local power grid.

At inference, though YOLOv11 and LaMa are designed efficient, it requires considerable computing resources for their real-time operation to process high rate frames maintaining good qualities of outputs. So when taken as a whole, this constant energy consumption over many deployments adds up and forms a large environmental footprint. Updates and precise retraining of models are also required to ensure precision and adaptivity, which increase the demand for energy. Another ignored element is the hardware manufacture and disposal impact, such as GPUs, cameras, and servers. Harmful pollution and resource depletion result from mining rare earth metals and during their production, and end-of-life waste management and recycling remain challenging for electronics. Still, the project has environmental upsides. For instance, automatic background-removal can minimize the requirement for video physical re-takes, studio setups, or on-location video shooting, saving on travel, and resource expenses. And the proposed system can help the environmental monitoring tasks in efficiently and accurately detecting objects or changes in natural scenes with less human interferences. To address environmental concerns, developers have a few options: first, they can use model pruning, quantization or take advantage of more energy-efficient hardware, like specialized AI accelerators.

Cloud vendors being encouraged to provide carbonneutral data centers, where such data centers can be preferentially used by applications/services for model training and serving. Tools for monitoring and reporting the energy footprint allow for a continual evaluation and improvement. As awareness increases, we're beginning to see sustainable AI practices take hold, focusing on a holistic lifecycle approach, from developing, deploying, and disposing of models. This is one more method to guarantee concerns of technology vs. environmental responsibility move forward in seeking ecological harmony while servicing computations. In addition, collaboration between policymakers, researchers, and industry stakeholders will be essential to set standards that encourage sustainable AI practices across the lifecycle of such systems. Transparent reporting of carbon usage during training and deployment can help

organizations stay accountable and push towards greener innovation. Ultimately, embedding environmental impact assessments into the development pipeline ensures that advancements in AI do not come at the expense of long-term ecological well-being.

5.7.1 Impact of the Work on Sustainability

The project's influence on sustainability is complicated, comprising challenges and potentials for fulfilling sustainability goals. On the one hand, the application of AI models such as YOLOv11 and LaMa for real-time automatic background person removal is computationally intensive. The carbon costs of training and running deep neural networks are high, and cannot be reversed unless managed through responsible usage. Such levels of energy consumption pose a serious threat to sustainable production, especially when run from non-renewable sources of energy, and serve as a stark reminder of the ever-present tension between development and environmental conservation.

But the project also presents possibilities for sustainability dividends when scaled sensibly into larger systems. The automation of those boring tasks, like removing the background from a video or photo, decreases the dependency on staff labor to do the manual lifting required in editing and reduce resource use and waste generated in traditional media production workflows. Faster, more efficient processing could lead to less physical infrastructure, fewer re-shoots, and less demand for travel or equipment, indirectly mitigating the environmental footprint of related industries, like film, advertising, and e-commerce.

In addition, AI-powered image-processing solutions foster a more sustainable digital economy by allowing remote team collaboration and remote experiences that have the potential to reduce both resource use and CO2. For instance, such technologies enable inclusive participation, as in virtual meetups, AR applications, or representation, without environmental costs related to commuting or venue operation. They also add to the circular economy and reuse and recycle content with a lifespan of digital assets, making images reusable and modifiable.

For sustainable impact, energy-efficient and responsible AI development needs to be prioritized in project implementation. This involves using techniques like model compression, pruning, optimizing for inference, using energy-efficient hardware, and selecting renewable energy sources for data centers. Transparency of energy and carbon use fosters accountability and culture of sustainability in AI RD and the products that it powers. In addition to improving social sustainability through the incorporated ethical considerations, equitable access to technology, unbiased outcomes, and protecting privacy and consent.

This integrated approach builds trust and maintains the social warp as well as weft of environmental care. Ultimately, the real sustainability of the project lies in continuous assessment and optimization throughout the entire AI life cycle, covering data collection, model training, deployment, and end-of-life hardware management. Sustainable AI

application based on regulations and corporate responsibility can also transform challenge into opportunity, to show that AI's innovation and sustainability do not collide but are in fact compatible. With a rising importance of sustainable development in the tech domain, this project needs to be part of a responsible AI mindset that sustains the planet as well as human well-being, economic prosperity, and cultural resilience in the long term.

The effect of the project on sustainability is largely dependent on the performance and efficiency of AI models like YOLOv11 and LaMa. YOLOv11 delivers high-object-detection accuracy in object detection with architectural improvements such as transformer-based attention and fine-tuned efficiency neck modules, while targeting competitive inference speeds. This increased efficiency lowers the computational costs during real-time operation which in turn reduces the energy consumed per operation compared to sub-optimized models. As a result, the model is suitable for practical use and can facilitate efficient use of resources for sustainability.

Furthermore, the fact that LaMa-based image inpainting is employed in the project has a large potential in the context of sustainable media production by enabling the automated execution of highly complex visual tasks which used to require a lot of costly human labor and material resources. By using the power of Libertas, creatives can be confident they won't have to fly out of town for a repeat shoot or a pickup shot, and that hours of time in the edit bay won't need to follow. The system's architecture also contributes to sustainable development through promoting remote collaboration and digital content generation, combating multiple forms of traditional carbon-intensive infrastructure and travel-based sources of greenhouse gas emissions.

With more sectors of industry transitioning into virtual worlds, this technology provides a foundation for a more sustainable approach to business – allowing effective, visually engaging remote interactions. Moreover, in order to move forward in a sustainable way, we also explore model compression, pruning, and quantization for YOLOv11 and LaMa to achieve even more lightweight and speed without losing performance. Running these optimized models on energy-efficient hardware, such as dedicated AI accelerators or renewable-powered data centers, could have a significant impact on the environment.

The whole perspective of sustainability goes beyond the above mentioned to social and economic dimensions as well. As it helps to break down cognitive and creative barriers to advanced visual processing technology, the tech can drive greater inclusivity and democratization of creative tools, benefit the planet, and empower communities all over the world to generate new ideas and find the solutions to today's challenges. The ability to create a better picture while using fewer resources really is a win-win when you consider the environmental implications of poor picture quality.

The problem of sustainability remains, however, particularly with respect to the life cycle of AI hardware. Material sourcing, e-waste management, responsible recycling, and

minimizing impact beyond operational emission are aspects that a project shall take into account. Integrating transparent reporting and carbon accounting into the project promotes accountability and encourages continuous improvement towards net-zero goals. Collaborative frameworks involving developers, users, policymakers, and environmental experts are essential to embed sustainability deeply in the technology's evolution.

Ultimately, this project embodies a microcosm of modern AI's dual role: driving transformative innovation while demanding conscientious stewardship to harmonize growth with planetary health. Embracing sustainability as a core design and operational principle can ensure that advanced AI technologies like YOLOv11 and LaMa contribute positively to a resilient, green future.

Building on the mental health implications of the work, any real-time background person removal solution that prioritizes privacy preservation, such as this project, has the potential to reduce the omnipresent sense of being looked upon in both private and public environments and its consequent mental stress and anxiety. Citizens who are aware that they are being watched feel vulnerable and tend to hold their tongues, reducing their sense of freedom and psychological health.

This technology, if used judiciously to blur or wipe out background people, could help counteract these negative psychological effects, contributing to a more agreeable digital space and making people feel valued and less on display. Furthermore, ethical use and consent are important to mitigate feelings of privacy violation, which may exacerbate against mental disorders such as anxiety or paranoia. Opening up the collection, processing, and security functions of data creates confidence rather than resistance or suspicion about surveillance technologies.

This trust is critical for personal well-being and for acceptance of AI-enabled systems to the benefit of society, particularly those designed to operate in sensitive domains such as schools, workplaces, or health care. The system's possibilities to accommodate health-related applications, such as tracking social distancing or health conditions using computer vision, portray the system's dual purpose that protects the public and sustains individuals' dignity. This baseline is important to reduce the social anxiety associated with invasive monitoring. Furthermore, the limited exposure of personal identifiers lessens the risk of identity theft and unauthorized profiling, which can have significant psychological impacts such as trauma and loss of control. The technology is enabling safer and transparent AI surveillance systems that can empower citizens and not make them feel as if they are being watched or threatened.

More largely, this project could impact privacy-preserving policies for next smart city infrastructures, which would incorporate the ethical AI guiding principles that prioritize mental health and human rights.

In addition, privacy-aware image processing can help create a more responsible relationship between individuals and automated technologies. The system can minimize unnecessary

exposure of visual information by processing only the content required for the intended task. This approach can reduce concerns about continuous identification and unnecessary collection of personal information. Clear communication about the purpose and limitations of the system can further improve users' understanding and confidence in AI-based applications. It is also important to ensure that automated processing does not unfairly discriminate against individuals because of differences in appearance, environment, or image quality. Continuous evaluation and testing can help identify potential biases and improve the reliability of the system in different real-world conditions. Organizations deploying such technologies should establish appropriate policies governing data access, retention, and responsible use. Human oversight should also remain an important component when automated image processing is applied in sensitive environments. These practices can help ensure that technological innovation supports human interests without unnecessarily compromising individual autonomy and privacy. Ultimately, privacy-preserving person removal can contribute to the development of AI systems that are more socially responsible, trustworthy, and acceptable to the wider community.

Positive effects include crowd control for incidents of mass gathering, anonymity while ensuring security. This helps people feel secure in mind, not just safe in body. Lastly, it puts a compelling example of practicing AI responsibly while maintaining individual freedom to it being the first-of-its-kind technology-wise.

5.8 Financial Analysis and Budget

The total work was done by author in the Kaggle environment. The T4 GPU was used which provides 30 hours per session in a week. This was sufficient to perform fine tuning on the pretrained model. The table 5.1 provides the total cost and budget of the overall work done till now.

Table 5.1: Financial Cost and Budget

Budget Components	**Estimated Cost (Taka)**
Software requirements	1000
Data acquisition	40,000
Computational resources	3000
Paid contents expense	1500
Miscellaneous	6000
Total estimated budget	**51,500**

CHAPTER VI

Addressing Complex Engineering Problems and Activities

5.1 Complex engineering problems associated with the current thesis

The automatic removal of background persons from camera frames raises a number of complex engineering issues. It can be difficult to design detection and segmentation mechanisms that accurately isolate target individuals while suppressing background people and irrelevant elements. Camera-captured frames are challenging, with variations in lighting, resolution, and scene complexity, necessitating robust algorithms that can consistently focus on the primary subject while filtering out unwanted persons. To be practical in real-world applications, these methods must also be scalable and efficient enough to process video streams in real time without sacrificing computational feasibility. Ensuring temporal consistency across consecutive frames is a significant challenge, as flickering or incomplete removal of background persons can degrade the user experience. The integration of multiple components introduces additional interdependence, where errors in one stage propagate to the others. Another challenge lies in balancing model accuracy with the hardware constraints of mobile and embedded devices, which often have limited computational and memory resources. Finally, privacy concerns, adaptability to diverse environments, and robustness against occlusions and motion blur add further layers of engineering complexity, making this problem space both technically demanding and socially relevant. Moreover, achieving low latency in processing is essential to maintain smooth and responsive user experiences, especially in interactive applications such as video conferencing or augmented reality. The dynamic nature of scenes, including rapid camera movements and occlusions, further complicates reliable person removal. Continuous adaptation and online learning mechanisms can enhance robustness by refining model predictions based on recent frames. Effective hardware acceleration and model optimization techniques are also vital to meet real-time constraints without compromising accuracy. Overall, addressing these diverse challenges requires a holistic, multidisciplinary approach combining advances in computer vision, hardware capabilities, and software engineering.

Table 5.2: A detailed description of Complex engineering problems associated with the thesis

Attribute	**Complex Engineering Problems**	
Range of conflicting requirements	EP2	Balancing the need for accurate detection and removal of background persons with the demand for real-time processing poses a fundamental challenge. The model must navigate tradeoffs between precision and efficiency while addressing diverse lighting conditions, camera resolutions, and scene complexities. Striking a balance between robustness against occlusions and computational feasibility adds another layer of conflicting requirements.
Depth of analysis required	EP3	This study requires a comprehensive and in-depth analytical approach. The analysis spans not only the technical aspects of the detection and inpainting pipeline but also extends to the broader implications of usability, privacy, and deployment in real-world camera-based applications. Achieving a thorough understanding of these multifaceted dimensions is essential for the proposed system and its application in automatic background person removal.
Familiarity of issues	EP4	Automatic background person removal introduces a set of recurring challenges. Understanding the complexities of dynamic backgrounds, the variability in camera frames, and the impact of imbalanced scenarios (e.g., crowded vs. sparse scenes) needs special awareness. Familiarity extends to the challenges of handling motion blur, lighting variations, and the integration of advanced computational methods. This depth of understanding is crucial in addressing the multifaceted issues inherent in background removal, ensuring the proposed model aligns with the practical demands of real-time vision systems.

Attribute	Complex Engineering Problems	
Extent of applicable codes	EP5	Understanding the extent of applicable codes is crucial, encompassing the utilization of advanced computer vision methods and innovative approaches for background person removal. The system's adaptability and implementation across diverse devices and environments underscore the wide-ranging applicability of the proposed codes.
Extent of stakeholder involvement and conflicting requirements	EP6	Background person removal systems involve multiple stakeholders, from end-users seeking privacy to developers aiming for efficient real-time pipelines. The need to meet conflicting requirements, such as accuracy, speed, and lightweight deployment on mobile devices, introduces further engineering complexity. Balancing privacy concerns, user expectations, and hardware limitations requires careful system design.
Interdependence	EP7	The overall architecture of the system, including object detection, segmentation, and inpainting components, showcases a carefully interconnected pipeline where each component relies on the others for optimal performance. This interdependence ensures that changes or improvements in one aspect can have ripple effects across the entire system, emphasizing the need for a holistic and integrated approach to addressing complex engineering challenges.

5.2 Complex engineering activities associated with the current thesis

Table 5.3: A detailed description of Complex engineering activities associated with the thesis

Attribute	**Addressing the Attributes of Complex Engineering Activitie**	
Range of resources	EA1	Automatic background person removal necessitates a broad range of resources. Peer-reviewed journal articles, open-source computer vision frameworks, public datasets (crowd scenes, surveillance footage, and real-world camera feeds), and computational resources for training deep learning models were utilized. Additionally, expertise in vision models, real-time processing, and system integration is crucial.
Level of interaction	EA2	Engaged with peers and researchers to exchange ideas and stay updated on advancements in real-time background removal and privacy-preserving vision. The iterative nature of model development requires constant feedback loops to refine the detection, segmentation, and inpainting pipeline. The interaction extends to the utilization of diverse datasets, incorporating both synthetic and real-world data, and addressing challenges in deployment. The high level of interaction reflects the complexity of coordinating multidisciplinary efforts in this engineering endeavor.

Attribute	Complex Engineering Activities	
Innovation	EA3	The proposed system introduces a hybrid pipeline combining state-of-the-art person detection (YOLO), segmentation (SAM), and inpainting (LaMa). This integration contributes to the advancement of real-time background removal, addressing challenges of accuracy, speed, and visual consistency. The research demonstrates a commitment to innovation, aiming to improve the efficiency and robustness of camera-based privacy solutions.
Consequences for society and the environment	EA4	The proposed background person removal system has significant societal benefits, enabling privacy protection, streamlining video processing workflows, and supporting applications in social media, surveillance, and AR/VR. It reduces risks of unwanted identity exposure and adapts to diverse scenarios. Environmentally, lightweight implementations reduce energy demands on mobile devices, aligning with sustainable computing goals.
Familiarity	EA5	The proposed system requires familiarity with real-time computer vision, person detection, segmentation, and deep learning architectures. Understanding video capture, optimization techniques, and deployment pipelines is crucial for effective implementation.

CHAPTER VII

Conclusions

5.1 Conclusion and Challenges Faced

The thesis work involved investigating the performance of different detection model and inpainting model. Choosing and utilizing the appropriate models to develop a pipeline that will perform background person detection and removal smoothly and accurately on a camera. The author faced many challenges during the thesis work. The challenges were collecting dataset of varied and complex background environment for proper learning of the chosen models so that it performs well on overall pipeline. Adapting and setting up each model for finetuning on the local pc of the author was also very challenging since each model requires different dependency on different devices. So setting up the appropriate dependency for each model to run properly was done overcoming many hurdles. In addition, optimizing the trade-off between accuracy and inference speed proved to be a critical factor, especially when considering real-time applications. The integration of the detection and inpainting modules into a seamless pipeline required careful adjustment to maintain synchronization and avoid latency.

Another major challenge was handling edge cases where occlusion, poor lighting, or low resolution significantly impacted the model performance. The thesis also highlighted the importance of proper evaluation metrics, as relying solely on accuracy was not sufficient to capture the practical effectiveness of the pipeline. Through experimentation, the author realized that continual model updates and domain-specific fine-tuning are necessary to sustain long-term performance. Moreover, ensuring that the system remains lightweight enough to run on modest hardware without compromising reliability was a key consideration.

Finally, the work provides a foundation for future research that can focus on improving model robustness, expanding dataset diversity, and exploring deployment in real-world environments. YOLOv11 includes sophisticated feature extraction module and a novel detection head that is more sensitive to objects at different scales, making it more capable of detecting and pinpointing objects like narrow and similar-sized crowd members. Its multi-use The faces of people who have been captured can be exploited without their consent. Detection models may also introduce biased results, leading to unbalanced treatment of specific demographic categories. Moreover, if not properly controlled, this technology can also expose surveillance

and abuse risks to privacy data of individuals. Openness, informed consent, and responsible data stewardship are essential to overcoming these challenges.

There's also the need to ensure fairness and accountability when it comes to model design and deployment. These ethical concerns must be addressed in the context of this area of research. And then there is the potential for such technologies to be used for mass surveillance, with insufficient legal or social oversight. Protecting privacy requires that data is anonymized and PII is limited. The ease at which deepfakes can be produced heightens the importance of ethical principles and regulation. Bias audits also should be considered as well as detectors to determine when unintentional discriminatory behavior arises with the system in daily operation. Involving ethicists, policy makers and the communities affected can help steer responsible development. Public understanding of the flaws and limitations.

Incorrect detection or incomplete removal could unintentionally alter important visual information within an image. Therefore, systematic testing should be performed using diverse images, environments, lighting conditions, and background structures. Such evaluation can help identify weaknesses that may not be visible when the system is tested only on controlled datasets.

Accountability mechanisms should be established so that users can report inappropriate results, privacy concerns, or unintended system behavior. Organizations should also define clear responsibilities for developers, administrators, and end users involved in deploying the technology. The collection and storage of training data should follow appropriate privacy and security practices.

Where possible, unnecessary personal information should be removed before data is used for training or evaluation. The system should not be deployed for purposes that conflict with applicable privacy regulations or the rights of individuals. Regular reviews can help ensure that the technology continues to comply with changing ethical standards and legal requirements. Independent assessments may also provide additional assurance regarding privacy protection, fairness, and system reliability. Public consultation can contribute valuable perspectives from communities that may be directly affected by automated image-processing technologies. Educational initiatives can further improve awareness of both the benefits and potential risks associated with AI-based systems. Developers should prioritize human oversight when the consequences of incorrect processing could be significant. Responsible deployment should therefore consider not only technical performance but also social, legal, and ethical implications. By incorporating these principles throughout the development lifecycle, potential risks can be identified and addressed at an early stage. Ultimately, fairness, transparency, privacy, and accountability should remain fundamental considerations in the development and deployment of AI-powered image processing systems.

5.2 Future Works

The thesis work has opened many future works. The result of the future work suggested promising directions for future investigations on real time inpainting. The overall pipeline was deployed on a local camera to perform the inpainting. More works can be done in future in deployment of the overall pipeline on any edge devices camera or surveillance system camera. Also the overall method can be implemented to perform the real time detection and inpainting on other other general objects along with persons for a generalized system,

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